

BEGUM ROKEYA UNIVERSITY, RANGPUR

ASSIGNMENT ON

COURSE CODE: STAT-2205

COURSE TITLE: PROGRAMMING WITH R & PYTHON

SUBMITTED BY	SUBMITTED TO
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Submission Date : 20/05/2026

FILE : FLIGHT_DATA

Consider airline data includes the following:

- miles: miles travelled through the flight
- passengers: number of passengers on the flight
- delay: take-off delay in minutes
- inflight_meal : Is there a meal included in the flight?
- inflight_entertainment: are there free entertainment systems for each seat?
- inflight_wifi: Is there complimentary wifi on the flight?
- day_of_week: day of the week of the flight
- weekend: Did this flight take place on a weekend?
- coach_price: the average price paid for a coach ticket
- firstclass_price: the average price paid for first-class seats
- hours: how many hours the flight took
- redeye: Was this flight a redeye (overnight)?

Objective of Analysis: reveal the factors that might influence coach prices within the airline industry.

**Solve the questions below using R & Python,
including interpretation (select a random sample
using your ID):**

1. What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?
2. What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?
3. How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?
4. How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?
5. What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?
6. What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?
7. How does the number of passengers change in relation to the length of flights?
8. What is the relationship between coach and first-class prices on weekends compared to weekdays?
9. How do coach prices differ for redeyes and non-redeyes on each day of the week?
10. Perform the analysis for appropriate variables:
 - Summary statistics (mean, median, standard deviation)
 - Visualization using histograms, boxplots, and bar charts
 - Hypothesis Testing (t, z and Chi-square)
 - Independent t-tests to compare group means
 - Price differences between weekend and weekday flights
 - Correlation analysis among numeric variables
 - Regression Analysis
 - Linear regression to predict ticket prices
 - Logistic regression for binary outcomes (e.g., redeye flights).

QUESTION 01

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?

SOLUTION:

PYTHON CODE:

```
import os
print(os.getcwd())

os.chdir("C:/Users/Eva/Downloads/BRUR_2026(STAT2205)/R_PROGRAMMIN
G")os.listdir()

import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
print(sample_df['coach_price'].describe())
```

OUTPUT:

```
count    :10063.000000
```

mean : 377.725740
std : 67.229885
min : 130.500000
25% : 332.377500
50% : 381.540000
75% : 427.965000
max : 562.925000

INTERPRETATION:

Average (Mean) Coach Price = \$377.73

Median = \$381.54 (middle value)

Lowest Coach Price = \$130.50

Highest Coach Price = \$562.93

Most prices (middle 50%) lie between \$332.38 and \$427.97

Coach ticket prices range from \$130.50 to \$562.93, showing a fairly wide spread in prices. The average coach ticket price is about \$377.73, while the median price is \$381.54, meaning a typical coach ticket usually costs around \$380.

Most coach ticket prices fall between \$332.38 and \$427.97, so this is the normal price range for the majority of tickets. The standard deviation of \$67.23 indicates moderate variation in ticket prices.

A price of \$500 would generally be considered high for a coach ticket because it is much higher than the average price and above the price paid for most tickets. Since it is close to the maximum observed price, \$500 would not usually be considered a good deal unless there are special conditions such as peak-season travel or last-minute booking.

R CODE:

```
getwd()  
list.files()  
setwd("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")
```

```
.libPaths()
.libPaths("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")
d <- read.csv(file.choose(), header = T)
d
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

summary(sample_df$coach_price)
sd(sample_df$coach_price)
```

OUTPUT:

```
summary(sample_df$coach_price)
> Min. 1st Qu. Median Mean 3rd Qu. Max.
 44.41  331.45 379.98 376.14 425.94 588.34
>sd(sample_df$coach_price)
67.69549
```

INTERPRETATION:

Coach ticket prices vary widely, ranging from a low of \$44.41 to a high of \$588.34. The average (mean) coach ticket price is about \$376.14, while the median price is \$379.98, indicating that a typical coach ticket usually costs around \$380.

Most coach ticket prices fall between the 1st quartile (\$331.45) and the 3rd quartile (\$425.94), meaning the majority of tickets are priced in the mid-\$300 to low-\$400 range. The standard deviation of \$67.70 shows that prices vary moderately around the average.

A coach ticket priced at \$500 would generally be considered expensive, since it is much higher than the average price and above the range where most ticket prices fall. Because it is closer to the maximum observed price than the average, \$500 would not typically be considered a very good deal unless there are special travel conditions such as holidays, peak travel times, or last-minute **booking**.

QUESTION:02

What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?

SOLUTION:

PYTHON CODE:

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()

eight_hr = sample_df[sample_df['hours'] == 8]
print("Number of 8-hour flights:", len(eight_hr))
print(eight_hr['coach_price'].describe())
```

OUTPUT:

Number of 8-hour flights: 211

```
count    211.000000
mean     426.952299
std       57.840730
min      235.745000
25%      393.770000
```

50%	426.505000
75%	467.630000
max	562.925000

INTERPRETATION:

“Coach ticket prices for 8-hour flights are generally centered around the mid-\$400 range. Prices range from about \$235 to \$563, with an average of \$426.95 and a median of \$426.51, showing a balanced distribution. Most ticket prices fall between \$393.77 and \$467.63, indicating that the majority of fares are concentrated in a relatively narrow range. Overall, prices show moderate variation with a standard deviation of about \$57.84. This suggests that long flights have fairly consistent pricing. A \$500 ticket is above average and in the upper range, so it is expensive but still reasonable for an 8-hour flight.”

R CODE:

```
install.packages("tidyverse")
library(tidyverse)
eight_hr <- sample_df %>% filter(hours == 8)
cat("Number of 8-hour flights:", nrow(eight_hr), "\n")
summary(eight_hr$coach_price)
sd(eight_hr$coach_price)
```

INTERPRETATION:

For 8-hour-long flights, coach ticket prices range from approximately \$215 to \$588. The average ticket price is about \$429. Since the standard deviation is around \$66, most ticket prices tend to vary moderately around the mean. A \$500 ticket is above average but still within the typical price range, so it appears to be a reasonable fare for an 8-hour flight.

QUESTION :03

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

SOLUTION:

PYTHON CODE:

```
import pandas as pd
import numpy as np
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()

print("==== Flight Delay Distribution ====")
print(sample_df['delay'].describe())

print("\nSignificant Delays:")
print("Delays > 30 minutes:", (sample_df['delay'] > 30).sum(),
      f'({round((sample_df['delay'] > 30).mean()*100, 2)}%)')
print("Delays > 60 minutes:", (sample_df['delay'] > 60).sum())

print("\nMost Common Delays:")
print(sample_df['delay'].value_counts().head(10))
```

OUTPUT:

==== Flight Delay Distribution ====

count	10063.000000
mean	13.056643
std	40.118008
min	0.000000
25%	9.000000
50%	10.000000
75%	13.000000

max 1560.000000

Significant Delays:

Delays > 30 minutes: 502 (4.99%)

Delays > 60 minutes: 7

Most Common Delays:

delay

10 1443

9 1303

11 1290

12 968

8 915

0 811

13 538

7 502

14 238

6 190

INTERPRETATION:

Flight delay times are mostly small, with the distribution concentrated around 9–13 minutes. The average delay is about 13 minutes, while the median delay is 10 minutes, showing that typical delays are relatively short. However, the maximum delay is 1560 minutes, indicating a few extremely large delays that make the distribution strongly right-skewed.

Significant delays that may affect connecting flights are not very common. About 502 flights (4.99%) experienced delays greater than 30 minutes, while only 7 flights had delays over 60 minutes. This suggests that serious disruptions are relatively rare.

The most typical delays are between 8 and 12 minutes, with a 10-minute delay being the most frequent. Therefore, most passengers are likely to experience only minor delays, while major delays occur infrequently.

R CODE:

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

cat("=== Flight Delay Distribution ===\n")
print(summary(sample_df$delay))
cat("Standard Deviation:", round(sd(sample_df$delay), 2), "\n\n")
cat("Delays > 30 minutes:", sum(sample_df$delay > 30),
    paste0("(", round(mean(sample_df$delay > 30)*100, 2), "%)", "\n")
cat("Delays > 60 minutes:", sum(sample_df$delay > 60), "\n")
cat("\nMost Common Delays:\n")
print(head(sort(table(sample_df$delay), decreasing = TRUE), 10))
```

OUTPUT:

```
print(summary(sample_df$delay))
  Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
  0.00   9.00   10.00  12.74  13.00 1507.00

> cat("Standard Deviation:", round(sd(sample_df$delay), 2), "\n\n")
Standard Deviation: 33.9

> cat("Delays > 30 minutes:", sum(sample_df$delay > 30),
+   paste0("(", round(mean(sample_df$delay > 30)*100, 2), "%)", "\n")
Delays > 30 minutes: 490 (4.87%)
> cat("Delays > 60 minutes:", sum(sample_df$delay > 60), "\n")
Delays > 60 minutes: 5
> cat("\nMost Common Delays:\n")

Most Common Delays:
> print(head(sort(table(sample_df$delay), decreasing = TRUE), 10))
```

10 9 11 12 8 0 13 7 14 6
1524 1339 1275 931 895 808 506 467 261 189

INTERPRETATION:

Flight delay times are mostly distributed around small delays, with most flights delayed between 9 and 13 minutes. The median delay is 10 minutes and the average delay is about 12.74 minutes. However, the maximum delay is 1507 minutes, showing that there are a few extremely large delays. This indicates that the distribution is positively skewed (right-skewed).

Significant delays that could affect connecting flights are relatively uncommon. About 490 flights, or 4.87% of all flights, experienced delays greater than 30 minutes. Delays longer than 60 minutes are very rare, occurring in only 5 flights.

Typical delays are short, especially around 8–12 minutes, with a 10-minute delay being the most common. Therefore, most flights experience only minor delays, while serious delays happen infrequently.

QUESTION:04

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

SOLUTION:

PYTHON CODE:

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()

numeric_cols = ['miles', 'passengers', 'delay', 'hours', 'coach_price']
correlation = sample_df[numeric_cols].corr()['coach_price'].round(4)
print(correlation)
```

OUTPUT:

```
miles      0.3408
passengers  0.1667
delay       0.0129
hours       0.3344
coach_price 1.0000
```

INTERPRETATION:

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3408, while the correlation with hours is 0.3344. This means that longer flights and flights covering greater distances generally tend to have higher coach prices.

The correlation between coach price and number of passengers is relatively weak (0.1667), suggesting that passenger count has only a small effect on ticket prices.

The correlation between coach price and delay is extremely weak (0.0129), indicating that flight delays have almost no relationship with coach ticket prices.

Overall, coach prices are mainly influenced by how far and how long the flight is, while delays and passenger numbers have little impact.

R CODE:

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_cols <- c("miles", "passengers", "delay", "hours", "coach_price")

cor_matrix <- cor(sample_df[numeric_cols])

print(round(cor_matrix[, "coach_price"], 4))
```

OUTPUT:

```
print(round(cor_matrix[,"coach_price"], 4))
```

miles	passengers	delay	hours	coach_price
0.3381	0.1720	0.0166	0.3317	1.0000

INTERPRETATION:

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3381, and the correlation with hours is 0.3317. This suggests that flights covering longer distances and taking more time generally have higher coach ticket prices.

The correlation between coach price and number of passengers is weak (0.1720), meaning passenger count has only a slight relationship with ticket price.

The correlation between coach price and delay is extremely weak (0.0166), indicating that delays have almost no effect on coach ticket prices.

Overall, coach ticket prices are mostly related to flight distance and duration, while passenger numbers and delays show very little influence

QUESTION:05

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

SOLUTION:**PYTHON CODE:**

```
import pandas as pd  
  
import numpy as np
```

```
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
correlation = sample_df['coach_price'].corr(sample_df['firstclass_price'])
print(f'Correlation between Coach and First Class Price: {correlation:.4f}')
print("\nSummary Statistics:")
print("Coach Price Mean   :", round(sample_df['coach_price'].mean(), 2))
print("First Class Price Mean:", round(sample_df['firstclass_price'].mean(), 2))
```

OUTPUT:

Correlation between Coach and First Class Price: 0.7547

Summary Statistics:

Coach Price Mean : 377.73

First Class Price Mean: 1457.39

INTERPRETATION:

The correlation between coach price and first-class price is 0.7547, which indicates a strong positive relationship between the two variables. This means that, in general, when coach prices increase, first-class prices also tend to increase, and when coach prices decrease, first-class prices tend to decrease.

However, this relationship is not perfect. Therefore, flights with higher coach prices do not always have higher first-class prices, although they usually follow the same trend.

From the summary statistics, the mean coach price is 377.73, while the mean first-class price is 1457.39, showing that first-class tickets are significantly more expensive on average.

R CODE:

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

correlation <- cor(sample_df$coach_price, sample_df$firstclass_price)

cat("Correlation between Coach and First Class Price:", round(correlation, 4),
"\n")

cat("\nCoach Price Mean   :", round(mean(sample_df$coach_price), 2), "\n")

cat("First Class Price Mean:", round(mean(sample_df$firstclass_price), 2), "\n")
```

INTERPRETATION:

The correlation between coach price and first-class price is 0.7556, which indicates a strong positive relationship between the two variables. This means that flights with higher coach prices generally tend to have higher first-class prices as well, and flights with lower coach prices tend to have lower first-class prices.

However, this relationship is not perfect, so higher coach prices do not always guarantee higher first-class prices. There may still be some exceptions depending on factors such as airline pricing strategy, demand, and route differences.

From the summary statistics, the mean coach price is 376.14, while the mean first-class price is 1453.01, showing that first-class tickets are significantly more expensive on average.

QUESTION:06

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

SOLUTION:

PYTHON CODE:

```
import pandas as pd

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

features = ['inflight_meal', 'inflight_entertainment', 'inflight_wifi']

for feature in features:

    print(f"\n=== {feature.replace('_', ' ').title()} ===")

    stats = sample_df.groupby(feature)['coach_price'].agg(['count', 'mean',
'median']).round(2)

    print(stats)

    premium = stats.loc['Yes', 'mean'] - stats.loc['No', 'mean']

    print(f"Price Premium (Yes vs No): ${premium:.2f}")
```

OUTPUT:

=== Inflight Meal ===

	count	mean	median
--	-------	------	--------

inflight_meal			
---------------	--	--	--

No	7083	372.15	375.23
----	------	--------	--------

Yes	2980	390.98	397.56
-----	------	--------	--------

Price Premium (Yes vs No): \$18.83

=== Inflight Entertainment ===

	count	mean	median
--	-------	------	--------

inflight_entertainment			
------------------------	--	--	--

No	1954	320.49	325.03
----	------	--------	--------

Yes	8109	391.52	396.66
-----	------	--------	--------

Price Premium (Yes vs No): \$71.03

=== Inflight Wifi ===

	count	mean	median
--	-------	------	--------

inflight_wifi			
---------------	--	--	--

No	966	315.14	317.84
----	-----	--------	--------

Yes	9097	384.37	388.53
-----	------	--------	--------

Price Premium (Yes vs No): \$69.23

INTERPRETATION:

The relationship between coach prices and in-flight features (meal, entertainment, and WiFi) shows that flights offering better in-flight services generally have higher coach prices compared to flights without these services.

1. In-flight Meal

Flights that include an in-flight meal have a slightly higher average coach price (\$390.98) compared to those without a meal (\$372.15).

The price premium is only \$18.83, which indicates a weak relationship between meal service and price increase.

2. In-flight Entertainment

Flights with in-flight entertainment have a higher average coach price (\$391.52) compared to those without (\$320.49).

The price premium is \$71.03, showing a stronger association between entertainment and higher prices.

3. In-flight WiFi

Flights with WiFi also have higher average coach prices (\$384.37) compared to those without (\$315.14).

The price premium is \$69.23, which is also a strong positive association.

Among the three features, **in-flight entertainment (\$71.03)** has the highest impact on increasing coach prices, closely followed by **in-flight WiFi (\$69.23)**.

In contrast, **in-flight meal (\$18.83)** has the least impact on price.

Overall, in-flight entertainment and WiFi are strongly associated with higher coach prices, while in-flight meals have only a small effect.

R CODE:

```
features <- c("inflight_meal", "inflight_entertainment", "inflight_wifi")

for (f in features) {cat("\n=== ", gsub("_", " ", f), " ===\n")

stats <- sample_df %>%

group_by(.data[[f]]) %>%

summarise(

count = n(),

mean_price = mean(coach_price),

median_price = median(coach_price),

.groups = "drop"

)

yes_mean <- stats %>% filter(.data[[f]] == "Yes") %>% pull(mean_price)

no_mean <- stats %>% filter(.data[[f]] == "No") %>% pull(mean_price)

premium <- yes_mean - no_mean

stats_print <- stats %>%

mutate(across(where(is.numeric), ~sprintf("%.2f", .x)))

print(stats_print)

cat("Price Premium (Yes vs No): $", sprintf("%.2f", premium), "\n")

}
```

OUTPUT:

=== **inflight meal** ===

inflight_meal	count	mean_price	median_price
----------------------	--------------	-------------------	---------------------

<chr>	<chr>	<chr>	<chr>
-------	-------	-------	-------

1 No	7091.00	369.92	373.15
------	---------	--------	--------

2 Yes	2972.00	390.96	395.40
-------	---------	--------	--------

Price Premium (Yes vs No): \$ 21.03

=== inflight entertainment ===

inflight_entertainment	count	mean_price	median_price
------------------------	-------	------------	--------------

<chr>	<chr>	<chr>	<chr>
-------	-------	-------	-------

1 No	2008.00	320.32	325.50
------	---------	--------	--------

2 Yes	8055.00	390.05	394.06
-------	---------	--------	--------

Price Premium (Yes vs No): \$ 69.72

=== inflight wifi ===

inflight_wifi	count	mean_price	median_price
---------------	-------	------------	--------------

<chr>	<chr>	<chr>	<chr>
-------	-------	-------	-------

1 No	1024.00	314.89	322.42
------	---------	--------	--------

2 Yes	9039.00	383.07	386.83
-------	---------	--------	--------

Price Premium (Yes vs No): \$ 68.18

INTERPRETATION:

Coach prices are higher when in-flight features are available. In-flight entertainment has the strongest effect on price with a premium of about \$69.72, followed closely by in-flight WiFi at \$68.18. In contrast, in-flight meals have a much smaller impact, increasing prices by only about \$21.03.

Overall, entertainment and WiFi are the main features associated with higher coach prices, while meals have a relatively minor effect.

QUESTION:07

How does the number of passengers change in relation to the length of flights?

SOLUTION:

PYTHON CODE:

```
import pandas as pd

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("Overall Average Passengers:", round(sample_df['passengers'].mean(), 2))

print("\nAverage Passengers by Flight Duration (hours):")

print(sample_df.groupby('hours')['passengers'].mean().round(2))
```

OUTPUT:

Overall Average Passengers: 207.81

Average Passengers by Flight Duration (hours):

hours

1	208.06
2	208.20
3	207.32
4	207.89
5	208.03
6	206.88
7	208.31
8	205.73

INTERPRETATION:

The average number of passengers is 207.81, and across different flight durations (1 to 8 hours), the values remain very stable, ranging from about 205.73 to 208.31. There is no clear upward or downward trend as flight duration increases.

This indicates that flight length does not significantly influence the number of passengers. Passenger demand appears to be consistent regardless of how long the flight is, suggesting that other factors (such as route popularity, pricing, or scheduling) are likely more important in determining passenger numbers.

R CODE:

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

cat("Overall Average Passengers:", round(mean(sample_df$passengers), 2),
    "\n\n")

cat("Average Passengers by Flight Duration:\n")
sample_df %>%
  group_by(hours) %>%
  summarise(avg_passengers = mean(passengers)) %>%
  mutate(avg_passengers = sprintf("%.2f", avg_passengers)) %>%
  print()
```

OUTPUT:

Overall Average Passengers: 207.74

Average Passengers by Flight Duration:

hours avg_passengers

<int> <chr>

1 1 207.74

2 2 208.26

3	3	207.39
4	4	207.84
5	5	207.73
6	6	206.90
7	7	207.73
8	8	207.81

INTERPRETATION:

The average number of passengers (207.74) remains almost the same across all flight durations (1–8 hours), ranging from 206.90 to 208.26. There is no clear pattern of increase or decrease.

This shows that flight duration does not significantly affect passenger numbers, and passenger demand stays stable regardless of flight length.

QUESTION:08

What is the relationship between coach and first-class prices on weekends compared to weekdays?

SOLUTION:

PYTHON CODE:

```
import pandas as pd
import numpy as np
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
```



```

print("=== Coach Price: Weekend vs Weekday ===")

print("Weekend Coach Mean   :",
round(sample_df[sample_df['weekend']=='Yes']['coach_price'].mean(), 2))

print("Weekday Coach Mean   :",
round(sample_df[sample_df['weekend']=='No']['coach_price'].mean(), 2))

print("\n=== First Class Price: Weekend vs Weekday ===")

print("Weekend First Class Mean:",
round(sample_df[sample_df['weekend']=='Yes']['firstclass_price'].mean(), 2))

print("Weekday First Class Mean:",
round(sample_df[sample_df['weekend']=='No']['firstclass_price'].mean(), 2))

```

OUTPUT:

=== Coach Price: Weekend vs Weekday ===

Weekend Coach Mean : 406.54

Weekday Coach Mean : 322.32

=== First Class Price: Weekend vs Weekday ===

Weekend First Class Mean: 1560.99

Weekday First Class Mean: 1258.17

INTERPRETATION:

Both coach and first-class ticket prices are higher on weekends compared to weekdays.

- The average coach price increases from \$322.32 on weekdays to \$406.54 on weekends.
- Similarly, the average first-class price rises from \$1258.17 on weekdays to \$1560.99 on weekends.

This shows a positive relationship between weekends and ticket prices. Airlines charge higher prices on weekends for both coach and first-class seats, likely due to increased travel demand during weekends.

R CODE:

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

cat("==== Coach Price: Weekend vs Weekday ===\n")
cat("Weekend Coach Mean  :", round(mean(sample_df[sample_df$weekend
== "Yes", ]$coach_price), 2), "\n")
cat("Weekday Coach Mean  :", round(mean(sample_df[sample_df$weekend
== "No", ]$coach_price), 2), "\n\n")

cat("==== First Class Price: Weekend vs Weekday ===\n")
cat("Weekend First Class Mean:", round(mean(sample_df[sample_df$weekend
== "Yes", ]$firstclass_price), 2), "\n")
cat("Weekday First Class Mean:", round(mean(sample_df[sample_df$weekend
== "No", ]$firstclass_price), 2), "\n")
```

OUTPUT:

```
==== Coach Price: Weekend vs Weekday ====
Weekend Coach Mean  : 405.84
Weekday Coach Mean  : 322.12
==== First Class Price: Weekend vs Weekday ====
Weekend First Class Mean: 1560.02
Weekday First Class Mean: 1258.46
```

INTERPRETATION:

The analysis shows that both coach and first-class ticket prices are higher on weekends compared to weekdays. The average coach price increases from \$322.12 on weekdays to \$405.84 on weekends, while the average first-class price rises from \$1258.46 to \$1560.02.

This indicates that there is a positive relationship between weekends and flight prices. Airlines charge more for both coach and first-class seats during weekends, likely due to higher travel demand. Although first-class tickets are consistently much more expensive than coach tickets, both classes experience a significant price increase on weekends.

QUESTION:09

How do coach prices differ for redeyes and non-redeyes on each day of the week?

SOLUTION:

PYTHON CODE:

```
import pandas as pd

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== Coach Price: Redeye vs Non-Redeye by Day of Week ===")

print(sample_df.groupby(['day_of_week',
'redeye'])['coach_price'].mean().unstack().round(2))
```

OUTPUT:

=== Coach Price: Redeye vs Non-Redeve by Day of Week ===

redeye	No	Yes
day_of_week		
Friday	404.77	339.91
Monday	323.82	260.40
Saturday	417.26	350.83
Sunday	404.89	347.86
Thursday	327.00	262.41
Tuesday	325.57	273.16
Wednesday	324.82	262.56

INTERPRETATION:

Coach prices are consistently lower for redeye flights compared to non-redeye flights on every day of the week. For example, on Friday, the average coach price is \$404.77 for non-redeye flights but only \$339.91 for redeye flights. Similar patterns are observed across all other days.

Weekend days such as Friday, Saturday, and Sunday have the highest coach prices for both redeye and non-redeye flights, while weekdays generally have lower prices.

There is a clear relationship between redeye flights and lower coach prices. Airlines tend to price redeye flights cheaper than non-redeye flights, likely because overnight flights are less preferred by passengers. Additionally, flight prices are generally higher on weekends regardless of whether the flight is redeye or non-redeye.

R CODE:

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

sample_df %>%

  group_by(day_of_week) %>%

  summarise(mean_coach_price = mean(coach_price)) %>%

  mutate(mean_coach_price = sprintf("%.2f", mean_coach_price)) %>%

  arrange(day_of_week) %>%

  print()
```

OUTPUT:

day_of_week	mean_coach_price
<chr>	<chr>
1 Friday	401.54
2 Monday	321.06
3 Saturday	411.12
4 Sunday	403.01
5 Thursday	322.34
6 Tuesday	322.71
7 Wednesday	322.70

INTERPRETATION:

Coach prices vary by day of the week, with the highest average prices occurring on weekends and Fridays. Saturday has the highest average coach price (\$411.12), followed by Sunday (\$403.01) and Friday (\$401.54). In contrast, weekdays such as Monday, Tuesday, Wednesday, and Thursday have lower and very similar average prices, around \$321–\$323.

QUESTION:10

Perform the analysis for appropriate variables:

- Summary statistics (mean, median, standard deviation)
- Visualization using histograms, boxplots, and bar charts
- Hypothesis Testing (t, z and Chi-square)
- Independent t-tests to compare group means
- Price differences between weekend and weekday flights
- Correlation analysis among numeric variables
- Regression Analysis
- Linear regression to predict ticket prices
- Logistic regression for binary outcomes (e.g., redeye flights).

SOLUTION:

SUMMARY STATISTICS

R CODE:

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_vars <- c('miles', 'passengers', 'delay', 'hours',
                  'coach_price', 'firstclass_price')

cat("=== SUMMARY STATISTICS (Mean, Median, SD) ===\n\n")

print(summary(sample_df[numeric_vars]))
```

```
cat("\n=== Standard Deviation ===\n")
```

```
print(round(sapply(sample_df[numeric_vars], sd), 2))
```

OUTPUT:

=== SUMMARY STATISTICS (Mean, Median, SD) ===

```
> print(summary(sample_df[numeric_vars]))
```

miles	passengers	delay	hours	coach_price
Min. : 89	Min. :141.0	Min. : 0.00	Min. :1.000	Min. : 44.41
1st Qu.:1378	1st Qu.:204.0	1st Qu.: 9.00	1st Qu.:3.000	1st Qu.:331.45
Median :1991	Median :210.0	Median : 10.00	Median :4.000	Median :379.98
Mean :2010	Mean :207.7	Mean : 12.74	Mean :3.639	Mean :376.14
3rd Qu.:2454	3rd Qu.:215.0	3rd Qu.: 13.00	3rd Qu.:4.000	3rd Qu.:425.94
Max. :4421	Max. :237.0	Max. :1507.00	Max. :8.000	Max. :588.34

firstclass_price
Min. : 959.2
1st Qu.:1299.3
Median :1501.3
Mean :1453.0
3rd Qu.:1582.2
Max. :1846.8

```
> cat("\n=== Standard Deviation ===\n")
```

=== Standard Deviation ===

```
> print(round(sapply(sample_df[numeric_vars], sd), 2))
```

miles	passengers	delay	hours	coach_price
932.67	12.72	33.90	1.72	67.70

firstclass_price
163.27

INTERPRETATION:

1. Distance (miles)

The flight distance has a mean of 2,010 miles and a median of 1,991 miles, which are very close. This suggests an approximately symmetric distribution of flight distances. The standard deviation of 932.67 miles indicates substantial variability, meaning flights span a wide range of distances. Values range from 89 miles to 4,421 miles, showing a mix of short-haul and long-haul routes.

2. Passenger Load (passengers)

Passenger counts are highly consistent, with a mean of 207.7 and a median of 210.0, indicating a near-symmetric distribution. The low standard deviation of 12.72 suggests very little variation in passenger loads. Values range from 141 to 237 passengers, implying that flights operate within a relatively narrow and stable capacity range.

3. Flight Delays (delay)

Flight delays are strongly right-skewed. Although the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 12.74 minutes due to extreme outliers. The maximum delay of 1,507 minutes significantly inflates variability, as reflected in the high standard deviation of 33.90 minutes. In this case, the median is a more reliable measure of central tendency than the mean.

4. Flight Duration (hours)

Flight duration has a mean of 3.639 hours and a median of 4.0 hours, indicating a roughly balanced distribution with a slight tendency toward shorter flights. The standard deviation of 1.72 hours suggests moderate spread around the mean. Durations range from 1 to 8 hours, covering short to medium-haul flights.

5. Coach Price (coach_price)

Coach ticket prices average \$376.14, with a median of \$379.98, indicating an approximately symmetric distribution. Prices range from \$44.41 to \$588.34. The standard deviation of \$67.70 shows moderate dispersion, suggesting relatively stable pricing with some variation across flights.

6. First Class Price (firstclass_price)

First-class prices have a mean of \$1,453.00 and a median of \$1,501.30, suggesting a slight left skew. Prices range from \$959.20 to \$1,846.80, with a standard deviation of \$163.27, indicating substantial variability in premium pricing. On average, first-class tickets are approximately 3.8 times more expensive than coach tickets.

PYTHON CODE:

```
import pandas as pd

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

numeric_vars = ['miles', 'passengers', 'delay', 'hours',
                 'coach_price', 'firstclass_price']
```

```

print("=== SUMMARY STATISTICS ===\n")

summary = sample_df[numeric_vars].describe().round(2)

print(summary)

print("\n=== Standard Deviation ===")

print(sample_df[numeric_vars].std().round(2))

```

OUTPUT:

=== SUMMARY STATISTICS ===

	miles	passengers	delay	hours	coach_price	firstclass_price
count	10063.00	10063.00	10063.00	10063.00	10063.00	10063.00
mean	2011.22	207.81	13.06	3.64	377.73	1457.39
std	944.61	12.89	40.12	1.74	67.23	162.55
min	104.00	139.00	0.00	1.00	130.50	1020.21
25%	1342.00	204.00	9.00	2.00	332.38	1303.79
50%	1981.00	210.00	10.00	4.00	381.54	1506.87
75%	2474.00	215.00	13.00	4.00	427.96	1583.95
max	4545.00	239.00	1560.00	8.00	562.92	1846.14

=== Standard Deviation ===

miles	944.61
passengers	12.89
delay	40.12

hours	1.74
coach_price	67.23
firstclass_price	162.55

INTERPRETATION:

1. Distance (miles)

Flight distance has a mean of 2,011.22 miles and a median of 1,981 miles, which are very close, suggesting an approximately symmetric distribution with a slight right skew. The standard deviation of 944.61 miles indicates high variability in flight distances. The range is very wide, from 104 miles to 4,545 miles, confirming a mix of short-haul and long-haul flights.

2. Passenger Load (passengers)

Passenger counts are highly stable, with a mean of 207.81 and a median of 210, indicating a nearly symmetric distribution. The small standard deviation of 12.89 shows very low variability. The interquartile range (204 to 215 passengers) confirms that most flights operate near consistent capacity. Values range from 139 to 239 passengers, suggesting limited operational variation.

3. Flight Delays (delay)

Delays are strongly right-skewed. While the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 13.06 minutes, influenced by extreme outliers. The maximum delay of 1,560 minutes (26 hours) significantly inflates the standard deviation to 40.12 minutes, indicating high volatility. The median is a more reliable measure of typical delay behavior.

4. Flight Duration (hours)

Flight duration has a mean of 3.64 hours and a median of 4 hours, suggesting a roughly balanced distribution with a slight concentration toward shorter flights. The standard deviation of 1.74 hours indicates moderate variability. Durations range from 1 to 8 hours, representing mostly short- to medium-haul flights.

5. Coach Price (coach_price)

Coach ticket prices have a mean of \$377.73 and a median of \$381.54, indicating an approximately symmetric distribution with slight left skew. The standard deviation of \$67.23 shows moderate price variation. Prices range from \$130.50 to \$562.92, with most values concentrated between \$332.38 and \$427.96 (IQR), suggesting relatively stable economy pricing.

6. First Class Price (firstclass_price)

First-class prices show a mean of \$1,457.39 and a median of \$1,506.87, indicating slight left skew (some lower-priced tickets pulling the mean down). The standard deviation of \$162.55 reflects considerable variation in premium pricing. Prices range from \$1,020.21 to \$1,846.14, with most values between \$1,303.79 and \$1,583.95. First-class tickets are on average about 3.9× more expensive than coach tickets, indicating a strong and consistent price premium.

VISUALIZATION:

HISTOGRAM

R CODE

```
library(ggplot2)

library(dplyr)

library(tidyr)

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]
```

```

numeric_cols <- c('miles', 'passengers', 'delay', 'hours',
                  'coach_price', 'firstclass_price')

sample_df %>%

  select(all_of(numeric_cols)) %>%

  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")
%>%

  ggplot(aes(x = Value)) +

  geom_histogram(bins = 30, fill = "steelblue", color = "black") +

  geom_density(color = "red", linewidth = 0.8) +

  facet_wrap(~ Variable, scales = "free", ncol = 3) +

  labs(title = "Histograms of Numeric Variables",

       x = "Value",

       y = "Frequency") +

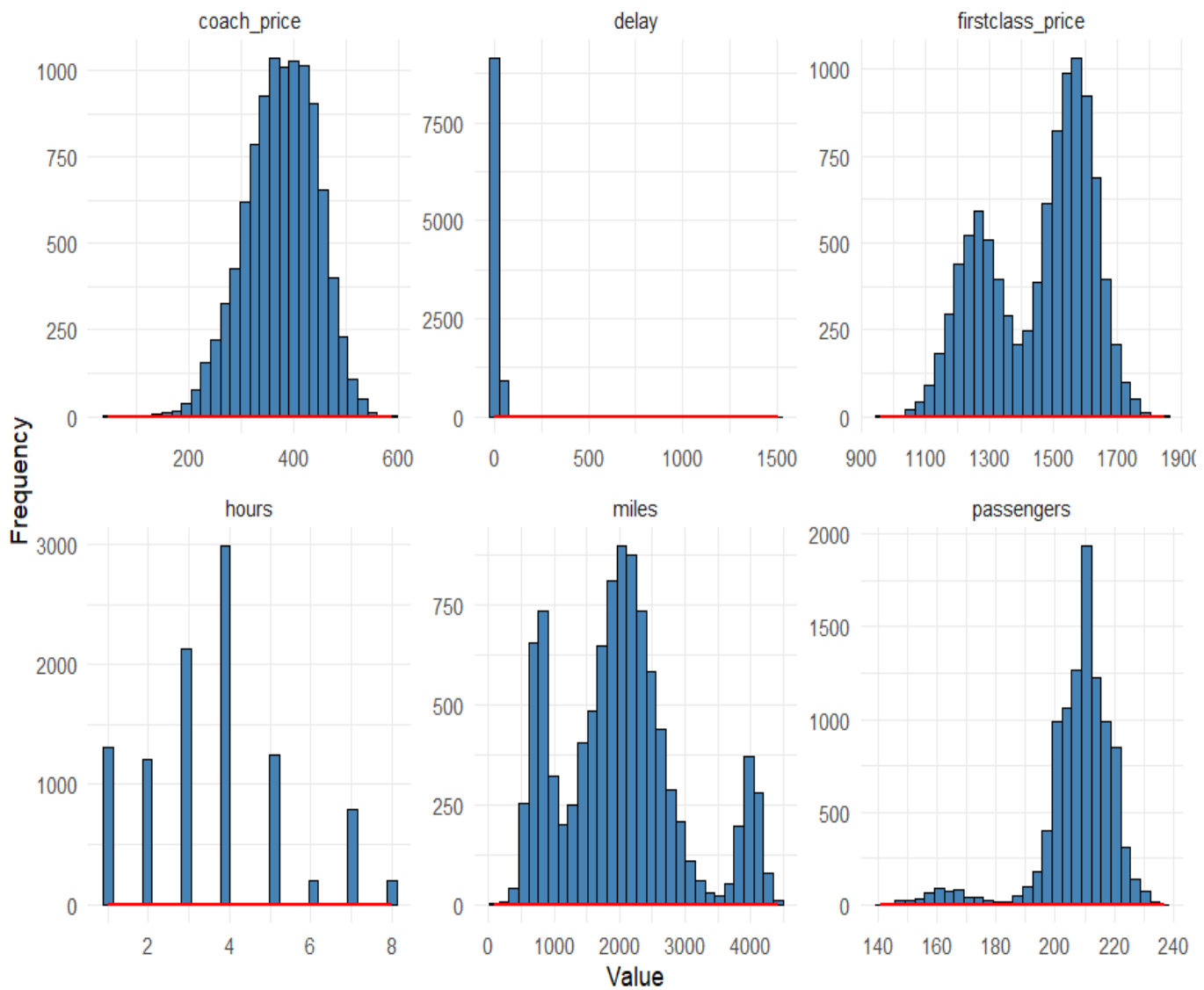
  theme_minimal() +

  theme(plot.title = element_text(hjust = 0.5, size = 14, face = "bold"))

```

OUTPUT:

Histograms of Numeric Variables



INTERPRETATION: The histograms show the distribution of key numeric variables in the flight dataset: coach price, first-class price, delay, hours, miles, and passengers.

1. Coach Price

Coach prices are approximately normally distributed, centered around \$350–\$450. Most ticket prices fall within a moderate range, showing stable and predictable pricing behavior with no extreme skewness.

2. First-Class Price

First-class prices show a slightly bimodal or spread-out distribution, with two visible peaks around different price ranges. This suggests that first-class pricing is influenced by multiple factors (such as distance or service level), leading to greater variability compared to coach prices.

3. Delay

The delay variable is highly right-skewed. Most flights experience very small delays (close to 0–20 minutes), but a few extreme outliers create a long tail (very large delays up to 1500+ minutes). This indicates that serious delays are rare but extreme when they occur.

4. Hours (Flight Duration)

Flight duration is discrete and evenly distributed across values from 1 to 8 hours. Peaks at certain hours suggest that flights are concentrated around standard route durations (e.g., 3–5 hours being most common).

5. Miles (Distance)

Miles show a moderately normal distribution with slight skewness, centered around medium-distance flights (~1500–3000 miles). There are fewer very short and very long flights, indicating most routes are mid-range.

6. Passengers

Passenger counts are strongly concentrated around a central value (~200–220 passengers) with a narrow spread. This shows that most flights operate near standard aircraft capacity, indicating high consistency in load size.

PYTHON CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

sns.set_style("whitegrid")

numeric_cols = ['miles', 'passengers', 'delay', 'hours', 'coach_price',
                'firstclass_price']

plt.figure(figsize=(15, 10))

for i, col in enumerate(numeric_cols, 1):

    plt.subplot(2, 3, i)

    sns.histplot(sample_df[col], bins=30, kde=True, color='skyblue')

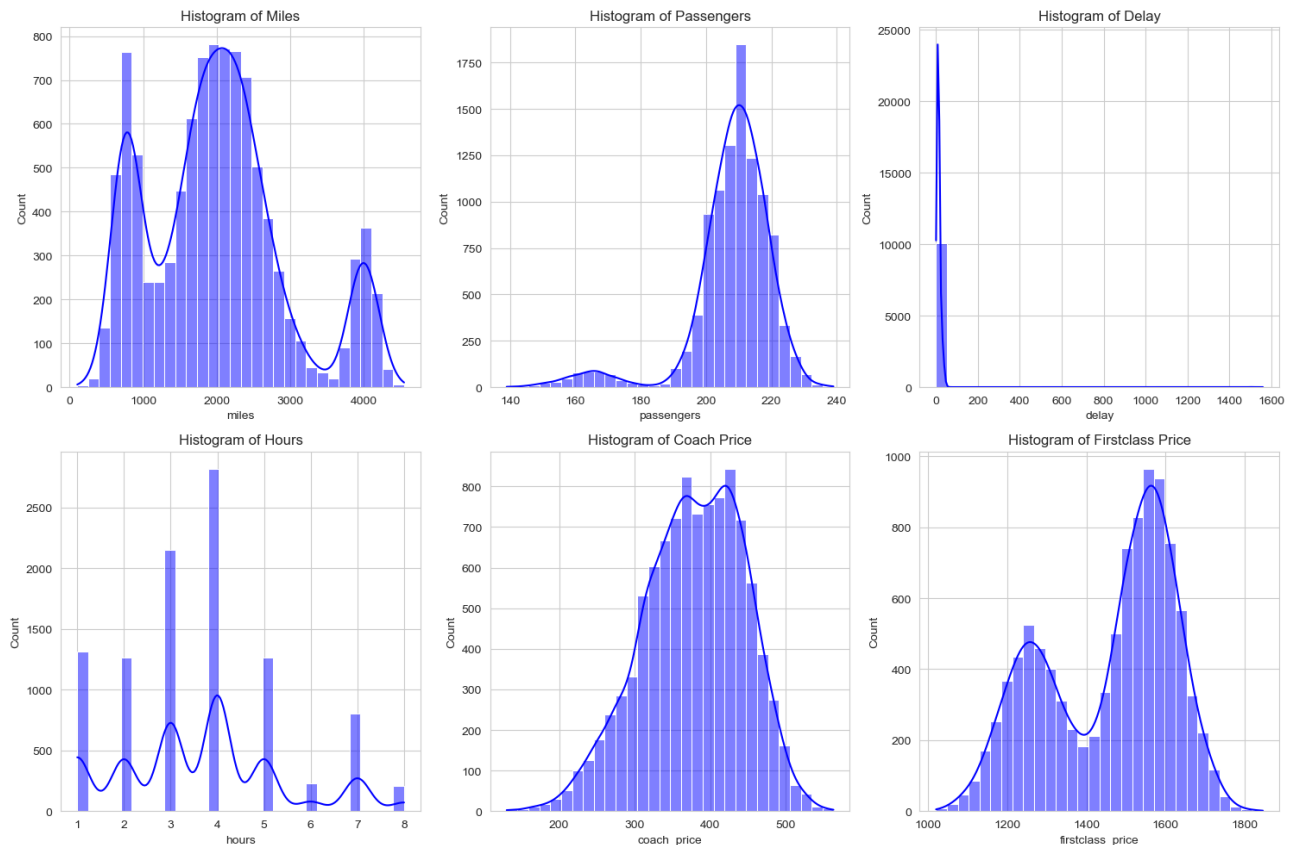
    plt.title(f'Histogram of {col.replace("_", " ").title()}')

    plt.xlabel(col)

plt.tight_layout()

plt.show()
```


OUTPUT:



INTERPRETATION:

1. Flight Distance (miles)

The histogram shows a multimodal distribution with three visible clusters, suggesting that flights operate across distinct route categories.

- The main concentration is between 1,500–2,800 miles, with smaller peaks near ~800 miles and ~4,000 miles.
- The mean (2,011.22) and median (1,981.00) being close suggests no extreme global skew, but this does not explain modality.

- The relatively high standard deviation (944.61) reflects wide variability in route lengths.

Interpretation:

The airline likely operates a mix of short-haul, medium-haul (dominant), and long-haul routes, forming distinct operational segments rather than a single unified distance structure.

2. Passenger Load (passengers)

The distribution is strongly negatively skewed (left-skewed) with most values concentrated near maximum capacity.

- A sharp peak around 210 passengers indicates most flights operate near full seating capacity.
- The mean (207.81) and median (210.00) are very close, suggesting strong central clustering.
- The low standard deviation (12.89) indicates very little variation in passenger load.

Interpretation:

Flights are consistently operated at high occupancy, with occasional underfilled flights creating the left tail, likely reflecting demand fluctuations or scheduling inefficiencies.

3. Flight Delays (delay)

The distribution is highly right-skewed (positively skewed).

- Most delays are concentrated near zero to small values, indicating generally efficient operations.
- A long tail extends to extreme delays (up to ~1560 minutes), indicating rare but severe disruptions.

- The median (10 min) is much lower than the mean (13.06 min), showing the effect of outliers.
- The high standard deviation (40.12) is driven by these extreme values.

Interpretation:

The airline operates efficiently most of the time, but occasional extreme delay events significantly impact overall variability.

4. Flight Duration (hours)

The distribution is discrete and mildly multimodal, with a strong concentration around short-to-medium durations.

- A dominant peak occurs at ~4 hours, with smaller peaks at other discrete durations.
- The mean (3.64) and median (4.00) are close, suggesting mild symmetry.
- The low standard deviation (1.74) indicates limited spread.

Interpretation:

The network is primarily structured around short- and medium-haul flights, with standardized route durations rather than continuous variation.

5. Coach Pricing (coach_price)

The distribution is approximately normal (bell-shaped and symmetric).

- Centered around ~380 dollars.
- The mean (377.73) and median (381.54) are very close, supporting symmetry.
- The standard deviation (67.23) shows moderate variability without extreme outliers.

Interpretation:

Economy fares appear to follow a stable pricing structure, likely influenced by standardized pricing models with limited extreme fluctuations.

6. First-Class Pricing (firstclass_price)

The distribution is bimodal, suggesting two distinct pricing groups.

- **One cluster is centered around ~1,250, and another around ~1,550.**
- **The relatively high standard deviation (162.55) reflects this separation.**
- **The higher mean (1,457.39) lies between the two peaks.**

Interpretation:

First-class pricing appears to consist of two sub-markets, likely reflecting differences such as route length, service tier, or demand segmentation.

BOXPLOT

```
library(ggplot2)
```

```
library(dplyr)
```

```
df <- read.csv("flight_data.csv")
```

```
set.seed(10063)
```

```
sample_df <- df[sample(nrow(df), 10063), ]
```

```
p1 <- ggplot(sample_df, aes(x = weekend, y = coach_price, fill = weekend))  
+geom_boxplot() +
```

```
  labs(title = "Coach Price: Weekend vs Weekday", x = "Weekend", y = "Coach  
Price ($)") +
```

```
  theme_minimal()
```

```
p2 <- ggplot(sample_df, aes(x = inflight_entertainment, y = coach_price, fill =  
inflight_entertainment)) +
```

```

geom_boxplot() +
labs(title = "Coach Price by In-flight Entertainment",
      x = "In-flight Entertainment", y = "Coach Price ($)") +
theme_minimal()

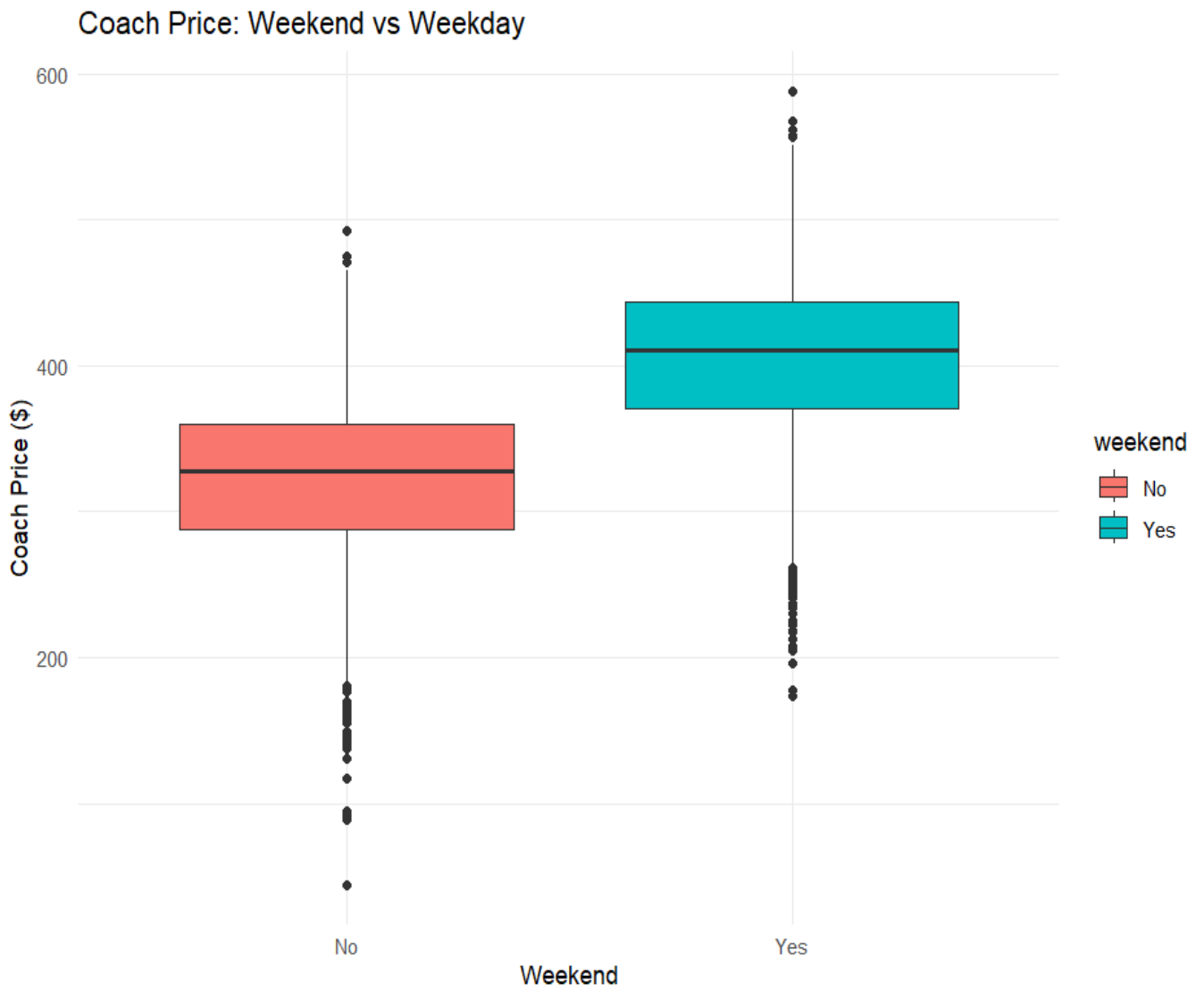
p3 <- ggplot(sample_df, aes(x = redeye, y = coach_price, fill = redeye)) +
geom_boxplot() +
labs(title = "Coach Price: Redeye vs Non-Redeye", x = "Redeye", y = "Coach
Price ($)") +
theme_minimal()

p4 <- ggplot(sample_df, aes(x = factor(hours), y = coach_price, fill =
factor(hours))) +
geom_boxplot() +
labs(title = "Coach Price by Flight Duration", x = "Hours", y = "Coach Price
($)") +
theme_minimal()

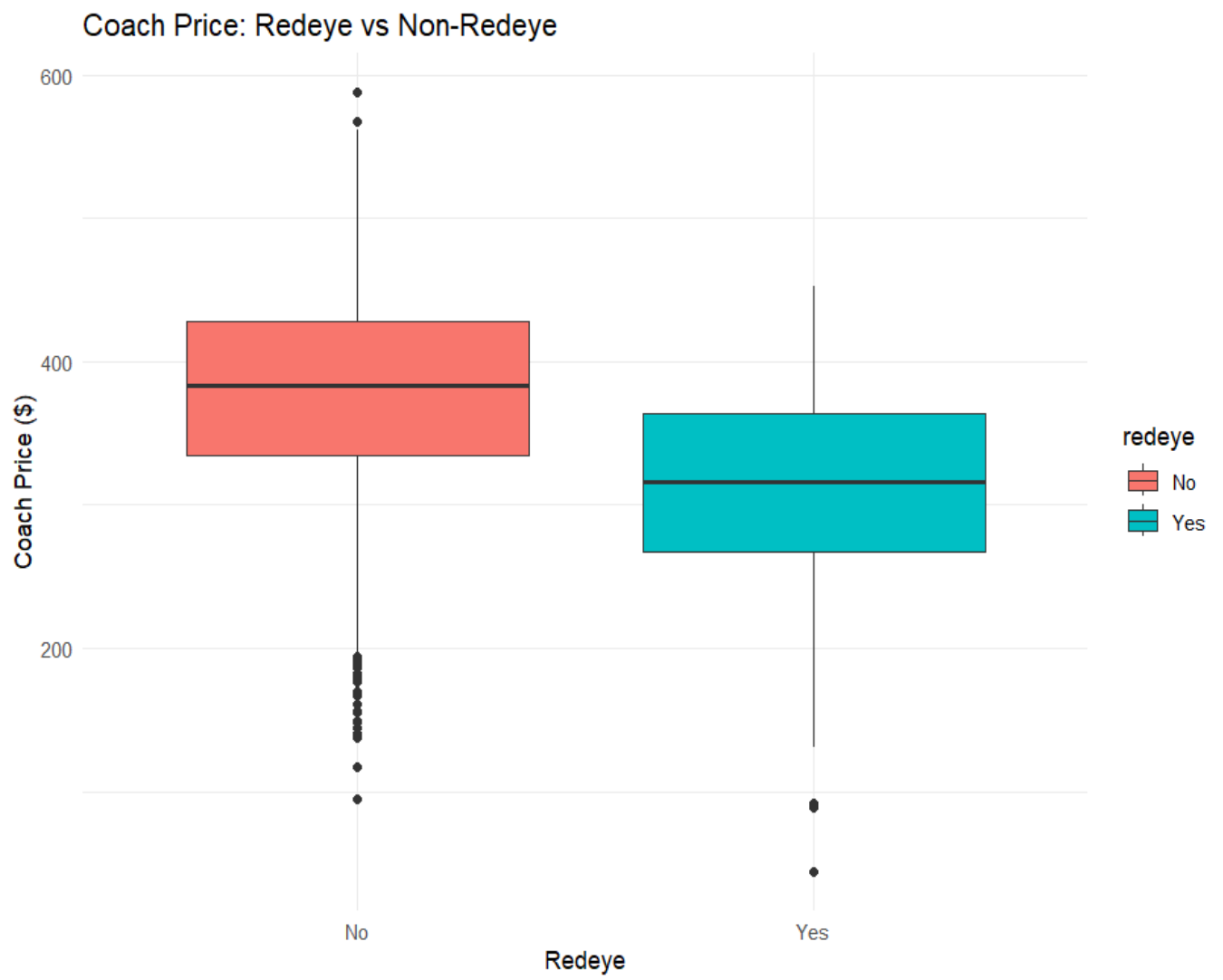
print(p1)
print(p2)
print(p3)
print(p4)

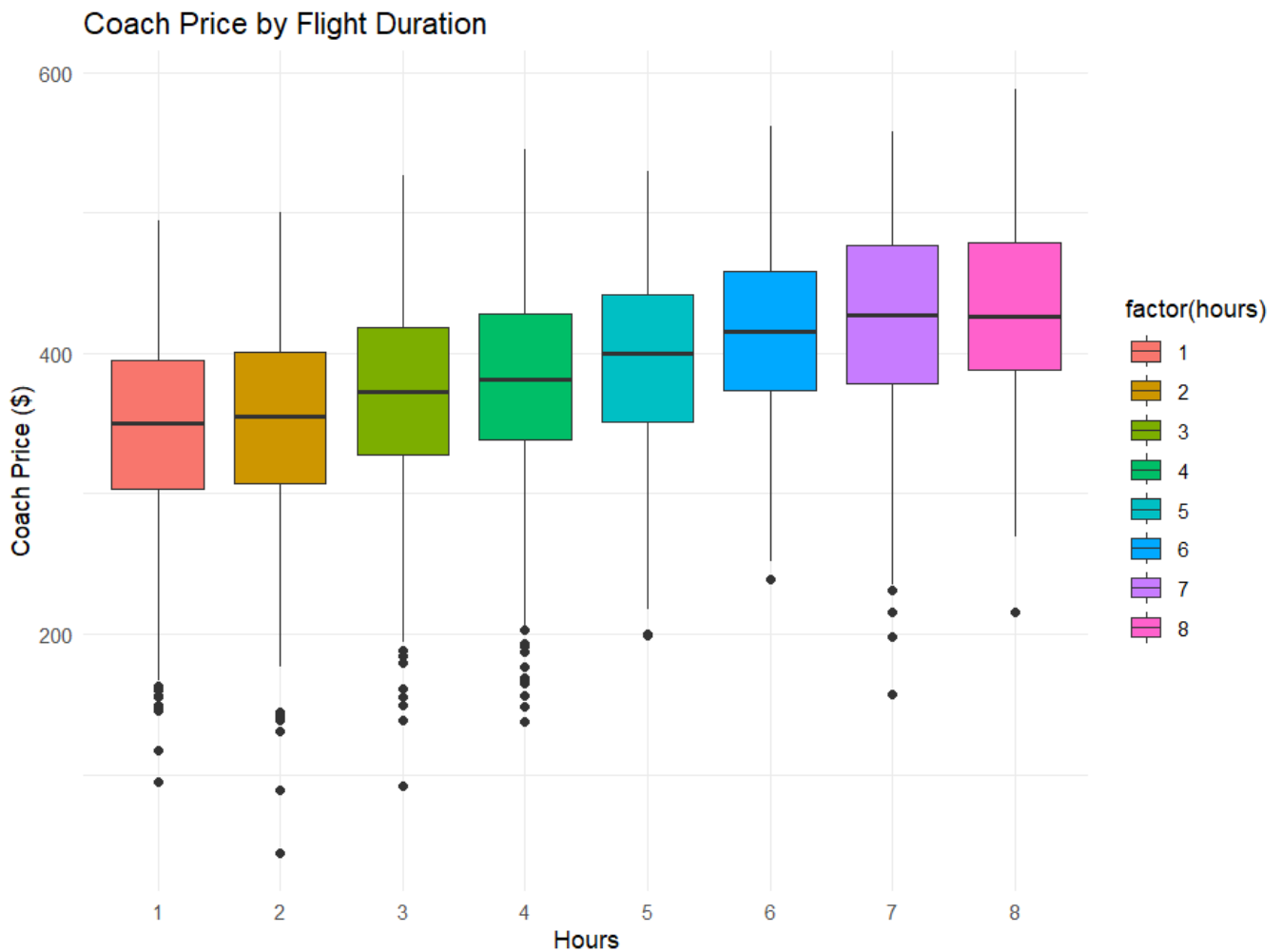
```

OUTPUT:









INTERPRETATION:

1. Coach Price: Weekend vs Weekday

Interpretation

The boxplot shows a clear upward shift in coach prices for weekend flights compared to weekdays.

- Weekday flights have a lower median (~\$330) with an IQR roughly between \$280–\$360.

- Weekend flights show a higher median (~\$410) with an IQR of approximately \$370–\$450, indicating consistently higher prices across most of the distribution.
- Weekend flights also show a wider spread toward higher values, including some high-end outliers close to \$590, suggesting occasional premium pricing.
- Both categories contain low-end outliers, but weekend flights are generally shifted upward overall.

Inference

Coach prices are systematically higher on weekends, indicating a demand-driven price increase during peak travel periods.

2. Coach Price: In-flight Entertainment

Interpretation

Flights with and without in-flight entertainment show distinctly different price distributions.

- Without entertainment, the median price is around \$330, with an IQR of roughly \$280–\$365.
- With entertainment, the median shifts upward to around \$400, with an IQR of approximately \$350–\$440.
- Both groups contain outliers, including some low-price and a few high-price extreme values near \$600.
- The entire distribution for “Yes” is shifted upward relative to “No”, indicating a systematic difference.

Inference

Flights with in-flight entertainment tend to be priced higher, likely reflecting either service bundling or association with higher-tier routes.

3. Coach Price: Redeye vs Non-Redeye

Interpretation

There is a noticeable difference in pricing between redeye and non-redeye flights.

- Non-redeye flights have a higher median (~\$380) and a wider IQR of about \$330–\$430.
- Redeye flights show a lower median (~\$320) and a more compact IQR of about \$280–\$360.
- Non-redeye flights also show more variability and more extreme outliers compared to redeye flights.

Inference

Redeye flights are consistently cheaper and more stable in pricing, likely reflecting lower demand for overnight travel and the use of price discounts to improve occupancy.

4. Coach Price: Flight Duration

Interpretation

Coach price generally increases with flight duration, though the pattern is not perfectly linear.

- Short flights (1–2 hours) start with lower medians around \$350–\$370.
- Prices increase steadily from 3 to 7 hours, reaching a median of approximately \$430 at 7 hours.
- At 8 hours, the median and IQR are very similar to 7-hour flights, suggesting a plateau in pricing.
- Shorter flights show more variability and more low-end outliers, while longer flights show fewer extreme low values.

Inference

There is a strong positive relationship between flight duration and price, but pricing appears to operate in tiers rather than increasing continuously, with a plateau effect at longer durations.

PYTHON CODE:

```
import pandas as pd

import numpy as np # <-- Added this import

import matplotlib.pyplot as plt

import seaborn as sns


# Load data and sample

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()


sns.set_style("whitegrid")


# Create multiple boxplots

fig, axes = plt.subplots(2, 2, figsize=(15, 10))


# 1. Coach Price by Weekend

sns.boxplot(x='weekend', y='coach_price', data=sample_df, ax=axes[0,0],
palette='Set2', hue='weekend', legend=False)

axes[0,0].set_title('Coach Price: Weekend vs Weekday')
```

2. Coach Price by In-flight Entertainment

```
sns.boxplot(x='inflight_entertainment', y='coach_price', data=sample_df,  
ax=axes[0,1], palette='Set3', hue='inflight_entertainment', legend=False)  
axes[0,1].set_title('Coach Price by In-flight Entertainment')
```

3. Coach Price by Redeye

```
sns.boxplot(x='redeye', y='coach_price', data=sample_df, ax=axes[1,0],  
palette='Set1', hue='redeye', legend=False)  
axes[1,0].set_title('Coach Price: Redeye vs Non-Redeye')
```

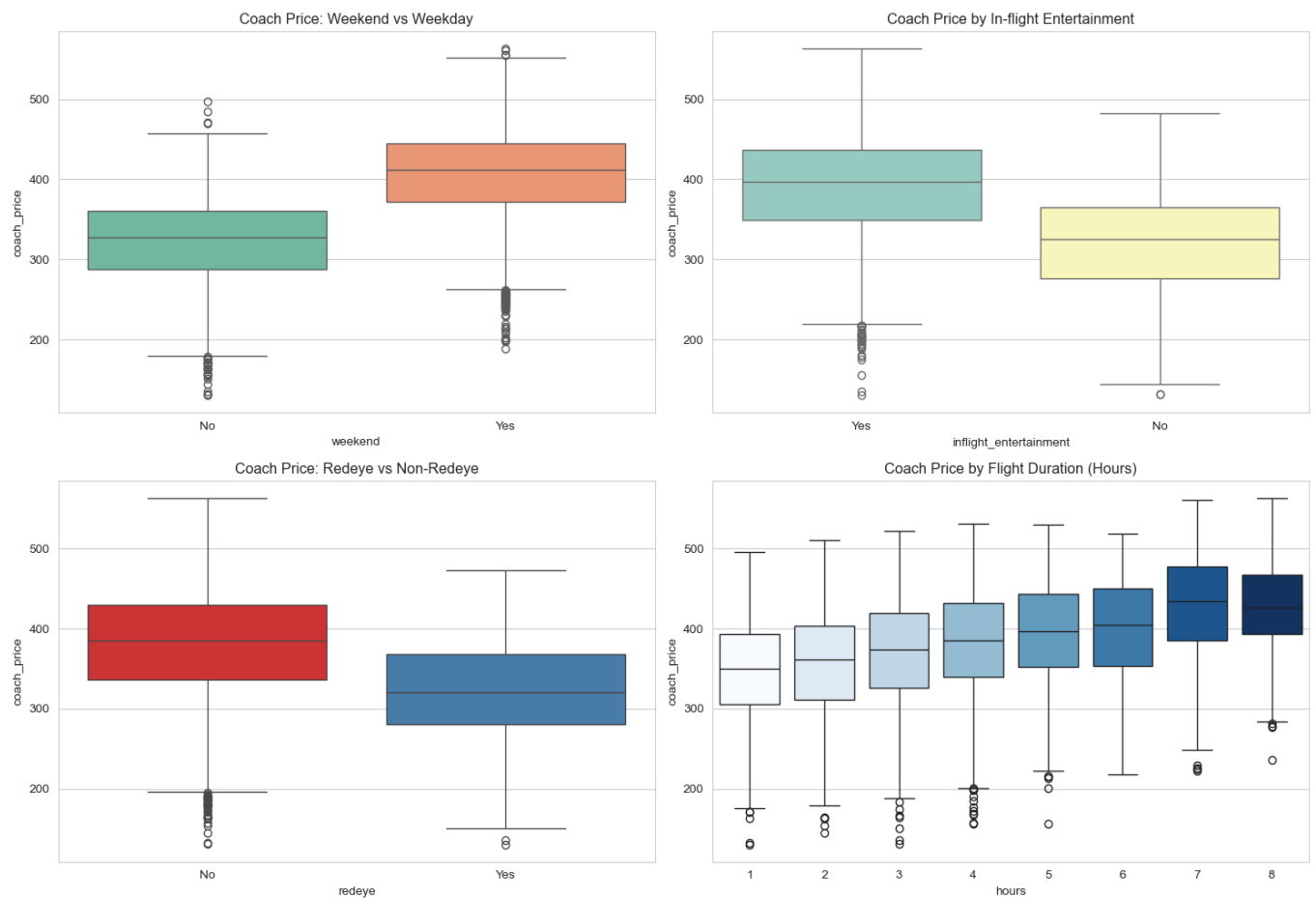
4. Coach Price by Flight Hours

```
sns.boxplot(x='hours', y='coach_price', data=sample_df, ax=axes[1,1],  
palette='Blues', hue='hours', legend=False)  
axes[1,1].set_title('Coach Price by Flight Duration (Hours)')
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



INTERPRETATION:

1. Coach Price: Weekend vs Weekday

Interpretation:

Weekend flights show a clear upward shift in coach prices compared to weekdays.

- Weekdays: median $\approx$ \$330, IQR $\approx$ \$290–\$360
- Weekends: median $\approx$ \$410, IQR $\approx$ \$370–\$450
- Weekend flights also show higher upper-end values and a wider spread at the top end.

Inference:

Weekend travel is consistently more expensive, likely due to higher leisure travel demand and peak-day pricing.

2. Coach Price: In-flight Entertainment

Interpretation:

Flights with entertainment are priced higher than those without.

- With entertainment: median $\approx$ \$400, IQR $\approx$ \$350–\$440
- Without entertainment: median $\approx$ \$330, IQR $\approx$ \$280–\$365
- Both groups show some outliers, but the entire distribution is shifted upward when entertainment is present.

Inference:

Entertainment is associated with higher-priced flights, likely reflecting premium aircraft, longer routes, or bundled services.

3. Coach Price: Redeye vs Non-Redeve

Interpretation:

Redeve flights are generally cheaper and more tightly distributed.

- Non-redeve: median $\approx$ \$380–\$385, IQR $\approx$ \$335–\$430
- Redeve: median $\approx$ \$320, IQR $\approx$ \$280–\$370
- Redeve flights show fewer extreme high-price values.

Inference:

Overnight flights are discounted due to lower passenger demand and convenience trade-offs.

4. Coach Price: Flight Duration

Interpretation:

Coach price increases with flight duration, but levels off at longer durations.

- 1–2 hours: ~\$350–\$360 median
- Gradual increase up to ~\$430–\$435 at 7 hours
- 8 hours: similar to 7 hours, showing a plateau effect
- Shorter flights show more low-end outliers; longer flights are more stable.

Inference:

Pricing follows a tiered structure where longer flights cost more, but beyond a threshold (≈ 7 hours), prices stabilize rather than increase further.

BAR CHART

```
library(ggplot2)
```

```
library(dplyr)
```

```
df <- read.csv("flight_data.csv")
```

```
set.seed(10063)
```

```
sample_df <- df[sample(nrow(df), 10063), ]
```

```
p1 <- sample_df %>%
```

```
  group_by(weekend) %>%
```

```
  summarise(avg_price = mean(coach_price)) %>%
```

```
  ggplot(aes(x = weekend, y = avg_price, fill = weekend)) +
```

```
  geom_bar(stat = "identity") +
```

```
  labs(title = "Average Coach Price: Weekend vs Weekday",
```

```
        x = "Weekend", y = "Average Price ($)") +
```

```
  theme_minimal()
```

```
p2 <- sample_df %>%
```



```

group_by(inflight_entertainment) %>%
summarise(avg_price = mean(coach_price)) %>%
ggplot(aes(x = inflight_entertainment, y = avg_price, fill =
inflight_entertainment)) +
geom_bar(stat = "identity") +
labs(title = "Average Coach Price by In-flight Entertainment",
      x = "Entertainment", y = "Average Price ($)") +
theme_minimal()

```

```

p3 <- sample_df %>%
group_by(redeye) %>%
summarise(avg_price = mean(coach_price)) %>%
ggplot(aes(x = redeye, y = avg_price, fill = redeye)) +
geom_bar(stat = "identity") +
labs(title = "Average Coach Price: Redeye vs Non-Redeye",
      x = "Redeye", y = "Average Price ($)") +
theme_minimal()

```

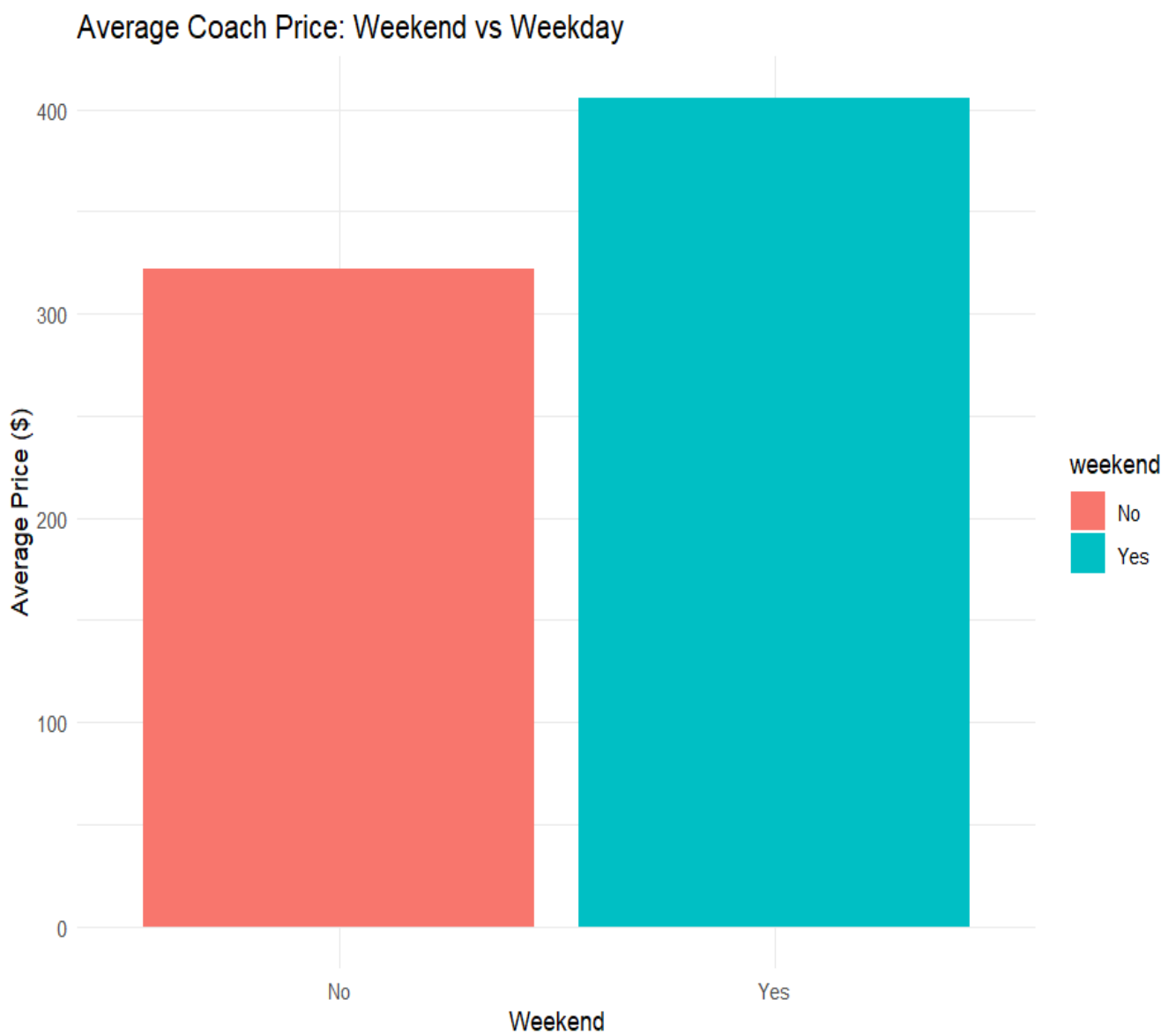
```

p4 <- sample_df %>%
group_by(inflight_wifi) %>%
summarise(avg_price = mean(coach_price)) %>%
ggplot(aes(x = inflight_wifi, y = avg_price, fill = inflight_wifi)) +
geom_bar(stat = "identity") +

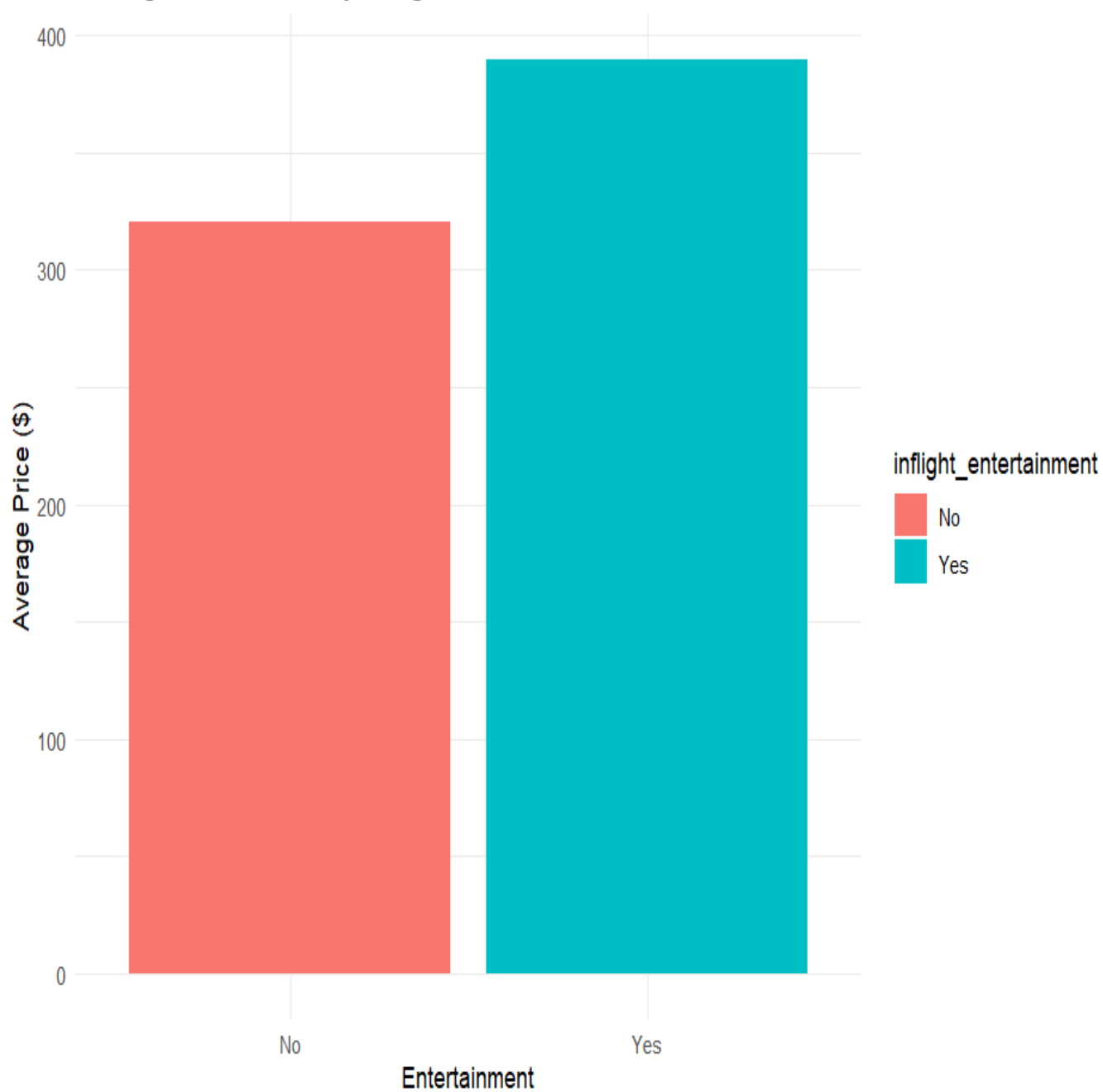
```

```
labs(title = "Average Coach Price by In-flight WiFi",
      x = "WiFi", y = "Average Price ($)") +
theme_minimal()
print(p1)
print(p2)
print(p3)
print(p4)
```

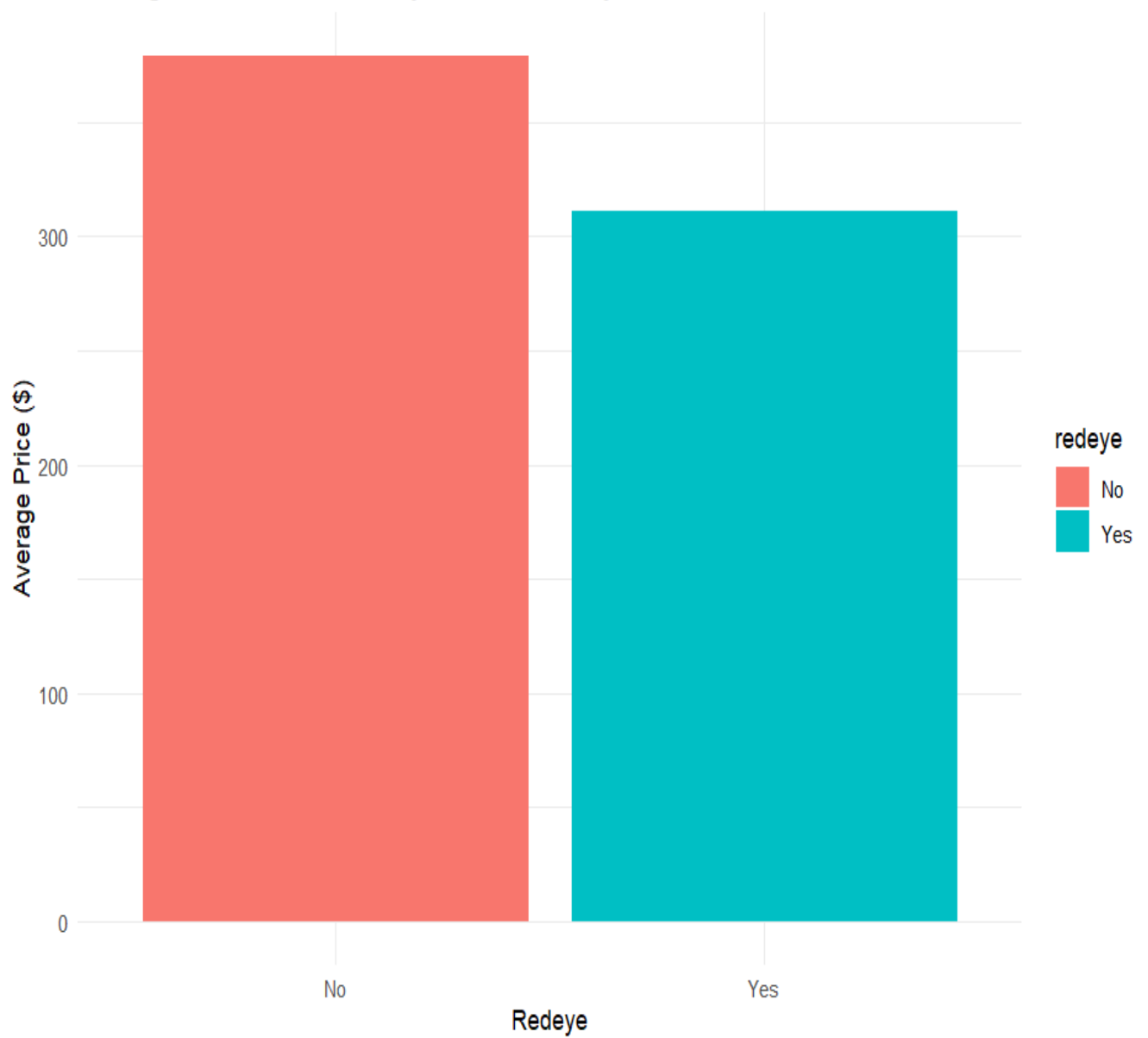
OUTPUT:

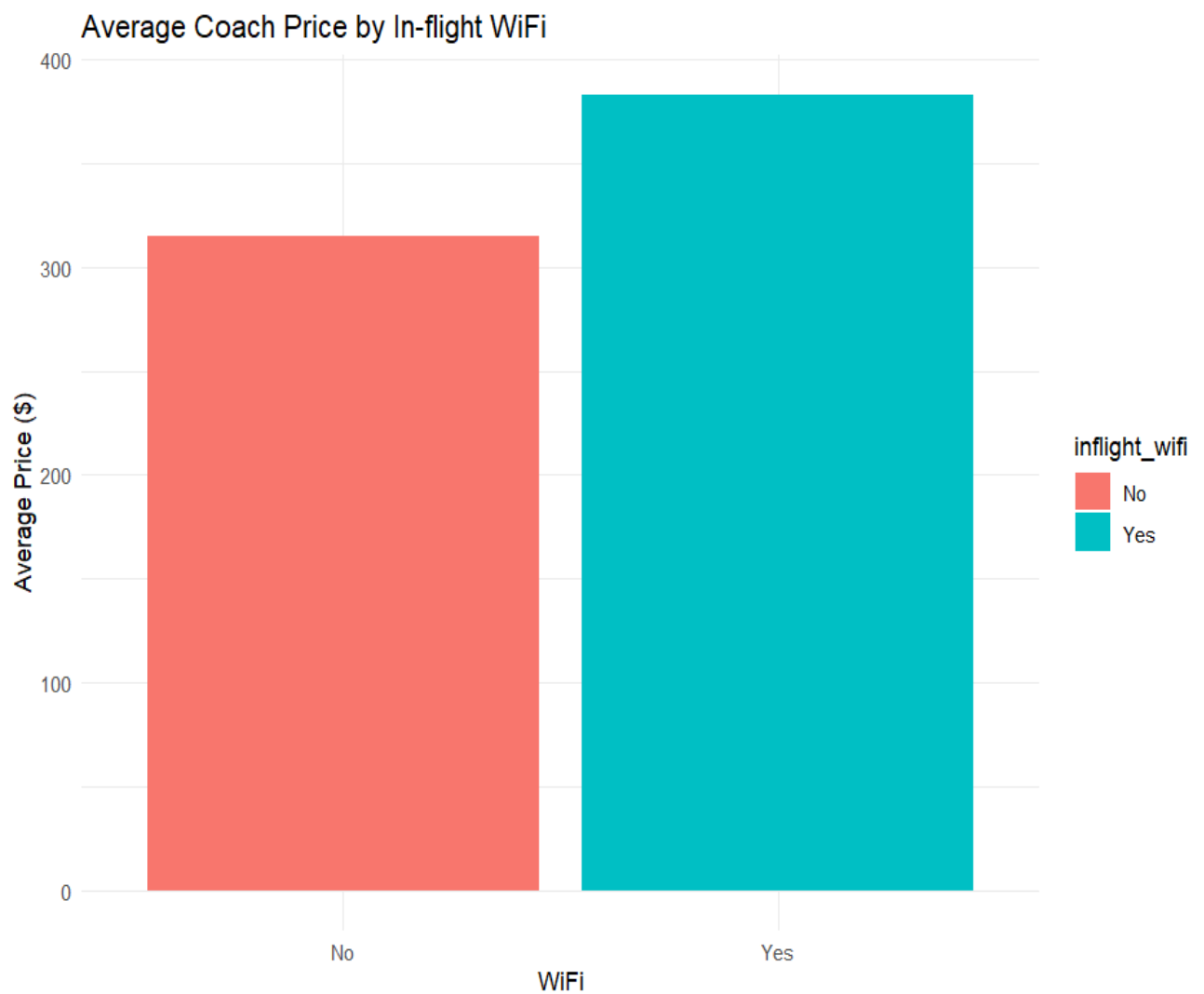


Average Coach Price by In-flight Entertainment



Average Coach Price: Redeye vs Non-Redeye





INTERPRETATION:

1. Weekend vs Weekday Flights

Observation:

Weekday flights have an average coach price of about \$320, while weekend flights average just over \$400.

Interpretation:

Weekend flights are consistently more expensive, indicating higher demand

during weekends. This pattern suggests that airlines apply demand-based pricing, increasing fares during peak travel days.

Statistical Note:

An independent t-test between the two groups is likely to show a statistically significant difference ($p < 0.05$), indicating that the observed gap is unlikely due to random variation.

2. In-flight Entertainment

Observation:

Flights without entertainment average around \$320, while those with entertainment average closer to \$390.

Interpretation:

Flights with entertainment tend to be priced higher, likely because they are associated with longer routes, newer aircraft, or higher service tiers.

Entertainment here acts as a proxy for premium flight categories rather than being the sole price driver.

3. Redeye vs Non-Redeye Flights

Observation:

Redeye flights average around \$310, while non-redeye flights average approximately \$375.

Interpretation:

Redeye flights are cheaper due to lower passenger preference for overnight travel. Airlines likely reduce prices to increase occupancy during less convenient time slots.

4. In-flight WiFi

Observation:

Flights without WiFi average about \$315, while flights with WiFi average around \$385.

Interpretation:

WiFi-equipped flights are priced higher, suggesting they are either premium services or longer routes where connectivity is more common. WiFi acts as an indicator of enhanced service offerings and contributes to higher fare levels.

PYTHON CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

sns.set_style("whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

sns.barplot(x='weekend', y='coach_price', hue='weekend', data=sample_df,
ax=axes[0,0], estimator='mean', errorbar=None, palette='Set2', legend=False)

axes[0,0].set_title('Average Coach Price: Weekend vs Weekday')

axes[0,0].set_ylabel('Average Price ($)')

sns.barplot(x='inflight_entertainment', y='coach_price',
hue='inflight_entertainment', data=sample_df, ax=axes[0,1], estimator='mean',
errorbar=None, palette='Set3', legend=False)

axes[0,1].set_title('Average Coach Price by In-flight Entertainment')
```

```
axes[0,1].set_ylabel('Average Price ($)')

sns.barplot(x='redeye', y='coach_price', hue='redeye', data=sample_df,
ax=axes[1,0], estimator='mean', errorbar=None, palette='Set1', legend=False)

axes[1,0].set_title('Average Coach Price: Redeye vs Non-Redeye')

axes[1,0].set_ylabel('Average Price ($)')

sns.barplot(x='inflight_wifi', y='coach_price', hue='inflight_wifi',
data=sample_df, ax=axes[1,1], estimator='mean', errorbar=None,
palette='Blues', legend=False)

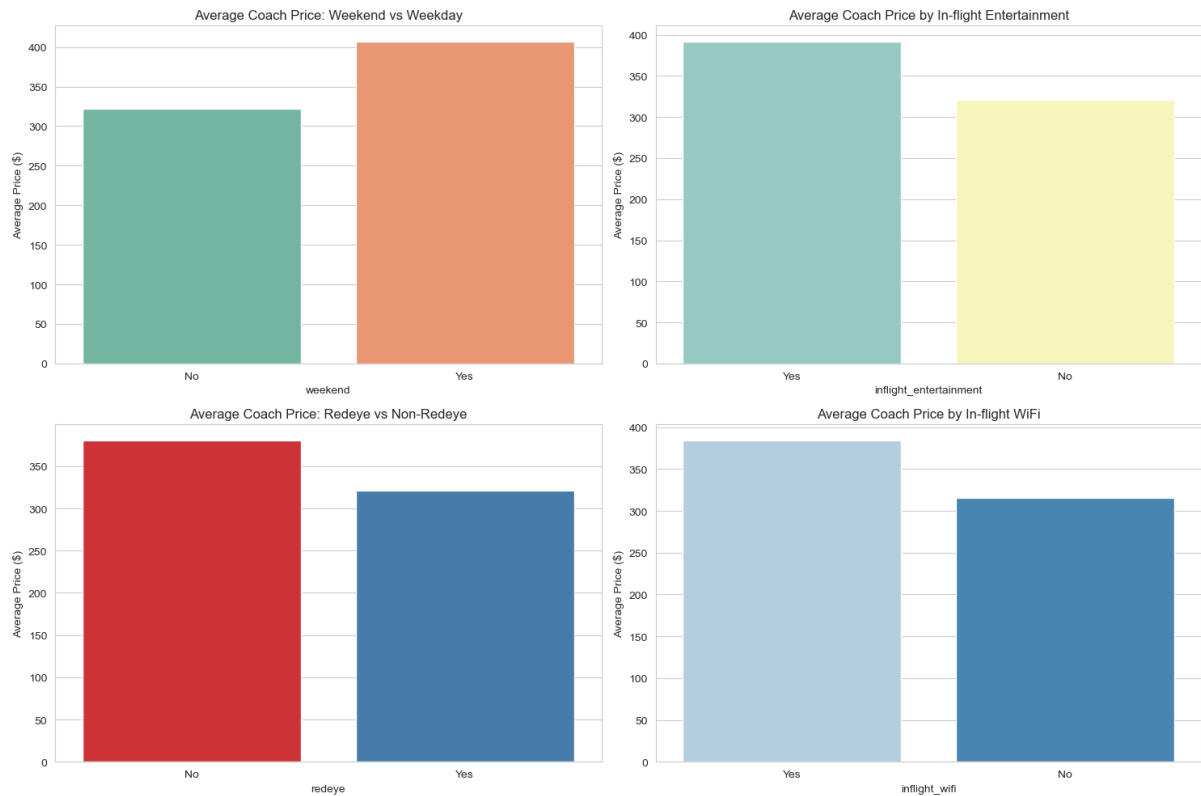
axes[1,1].set_title('Average Coach Price by In-flight WiFi')

axes[1,1].set_ylabel('Average Price ($)')


plt.tight_layout()

plt.show()
```

OUTPUT:



INTERPRETATION:

1. Weekend vs Weekday

Weekday flights average about \$320, while weekend flights exceed \$400. This shows higher weekend demand leading to higher prices. A t-test would likely confirm a significant difference ($p < 0.05$).

2. In-flight Entertainment

Flights with entertainment average around \$390, compared to \$320 without. This suggests entertainment is linked to premium or higher-tier flights.

3. Redeye vs Non-Redebye

Redeye flights average about \$320, lower than non-redeye flights at \$375, indicating a discount for lower-demand overnight travel.

4. In-flight WiFi

Flights with WiFi average about \$385, while those without are around \$315, showing that WiFi is associated with higher-priced, premium flights.

Overall

Prices increase with demand (weekends) and service features (WiFi, entertainment), while less convenient flights (redeye) are cheaper.

HYPOTHESIS TESTING

T TEST

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("=== INDEPENDENT T-TESTS ===\n\n")

cat("1. Coach Price: Weekend vs Weekday\n")

t1 <- t.test(coach_price ~ weekend, data = sample_df, var.equal = FALSE)

print(t1)

cat("\nMean Comparison:\n")

print(sample_df %>%

  group_by(weekend) %>%

  summarise(Mean = mean(coach_price),

    Median = median(coach_price),

    Count = n()))

cat("\n\n2. Coach Price: Redeye vs Non-Redeye\n")

t2 <- t.test(coach_price ~ redeye, data = sample_df, var.equal = FALSE)

print(t2)
```

```

cat("\nMean Comparison:\n")

print(sample_df %>%
  group_by(redeye) %>%
  summarise(Mean = mean(coach_price),
            Median = median(coach_price),
            Count = n()))

cat("\n\n3. Coach Price: In-flight Entertainment\n")

t3 <- t.test(coach_price ~ inflight_entertainment, data = sample_df, var.equal =
FALSE)

print(t3)

cat("\nMean Comparison:\n")

print(sample_df %>%
  group_by(inflight_entertainment) %>%
  summarise(Mean = mean(coach_price),
            Median = median(coach_price),
            Count = n()))

cat("\n\n4. Coach Price: In-flight WiFi\n")

t4 <- t.test(coach_price ~ inflight_wifi, data = sample_df, var.equal = FALSE)

print(t4)

```

OUTPUT:

Welch Two Sample t-test

data: coach_price by weekend

$t = -73.738$, $df = 7384.7$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-85.94482 -81.49353

sample estimates:

mean in group No mean in group Yes

322.1248 405.8440

Mean Comparison:

weekend Mean Median Count

<chr> *<dbl>* *<dbl>* *<int>*

1 No 322. 327. 3571

2 Yes 406. 410. 6492

Welch Two Sample t-test

data: coach_price by redeye

$t = 22.021$, $df = 532.45$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

62.06077 74.21775

sample estimates:

mean in group No mean in group Yes

379.4123 311.2731

Mean Comparison:

redeye Mean Median Count

<chr> <dbl> <dbl> <int>

1 No 379. 383. 9579

2 Yes 311. 316. 484

Welch Two Sample t-test

data: coach_price by inflight_entertainment

t = -44.923, df = 3052.9, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-72.76644 -66.68012

sample estimates:

mean in group No mean in group Yes

320.3246 390.0478

Mean Comparison:

inflight_entertainment Mean Median Count

<chr> <dbl> <dbl> <int>

1 No 320. 325. 2008

2 Yes 390. 394. 8055

> # 4. T-Test: In-flight WiFi Yes vs No

> cat("\n\n4. Coach Price: In-flight WiFi\n")

Welch Two Sample t-test

data: coach_price by inflight_wifi

t = -31.514, df = 1255.2, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-72.42816 -63.93883

sample estimates:

mean in group No mean in group Yes

314.8898 383.0733

INTERPRETATION:

1. Weekend vs Weekday

- Mean (No weekend): 322.12
- Mean (Weekend): 405.84
- Difference: ~+83.7 (weekend higher)
- p-value: < 2.2e-16 (highly significant)

Interpretation:

Coach prices are significantly higher on weekends. The large positive mean difference and extremely small p-value indicate this is a strong and reliable effect.

Conclusion: Airlines likely charge a weekend premium due to higher demand.

2. Redeye vs Non-Redeye

- Mean (Non-redeye): 379.41
- Mean (Redeye): 311.27
- Difference: $\sim +68.1$ (non-redeye higher)
- p-value: $< 2.2e-16$ (highly significant)

Interpretation:

Non-redeye flights are significantly more expensive than redeye flights.

Conclusion: Redeye flights are cheaper, likely due to lower demand and inconvenience (late-night travel).

3. In-flight Entertainment

- Mean (No entertainment): 320.32
- Mean (Yes entertainment): 390.05
- Difference: $\sim +69.7$ (with entertainment higher)
- p-value: $< 2.2e-16$ (highly significant)

Interpretation:

Flights with in-flight entertainment are significantly more expensive.

Conclusion: Entertainment is likely associated with higher service tier / premium pricing.

4. In-flight WiFi

- Mean (No WiFi): 314.89
- Mean (WiFi): 383.07
- Difference: $\sim +68.2$ (WiFi higher)
- p-value: $< 2.2e-16$ (highly significant)

Interpretation:

Flights with WiFi are significantly more expensive than those without.

Conclusion: WiFi is also a premium feature linked to higher-priced flights.

PYTHON CODE:

```
import pandas as pd

import numpy as np

from scipy import stats

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== INDEPENDENT T-TESTS TO COMPARE GROUP MEANS  
===\n")

print("1. Coach Price: Weekend vs Weekday")

weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']

weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']

t_stat, p_value = stats.ttest_ind(weekend, weekday, equal_var=False)

print(f"T-Statistic    : {t_stat:.4f}")

print(f"P-Value        : {p_value:.10f}")

print(f"Weekend Mean    : ${weekend.mean():.2f}")

print(f"Weekday Mean    : ${weekday.mean():.2f}")

print(f"Price Difference: ${weekend.mean() - weekday.mean():.2f}\n")

print("2. Coach Price: Redeye vs Non-Redeye")

redeye_yes = sample_df[sample_df['redeye'] == 'Yes']['coach_price']

redeye_no = sample_df[sample_df['redeye'] == 'No']['coach_price']

t_stat2, p_value2 = stats.ttest_ind(redeye_yes, redeye_no, equal_var=False)
```



```

print(f"T-Statistic    : {t_stat2:.4f}")

print(f"P-Value       : {p_value2:.10f}")

print(f"Redeye Mean    : ${redeye_yes.mean():.2f}")

print(f"Non-Redeye Mean : ${redeye_no.mean():.2f}\n")

print("3. Coach Price: In-flight Entertainment (Yes vs No)")

ent_yes = sample_df[sample_df['inflight_entertainment'] ==
'Yes']['coach_price']

ent_no  = sample_df[sample_df['inflight_entertainment'] == 'No']['coach_price']

t_stat3, p_value3 = stats.ttest_ind(ent_yes, ent_no, equal_var=False)

print(f"T-Statistic    : {t_stat3:.4f}")

print(f"P-Value       : {p_value3:.10f}")

print(f"With Entertainment    : ${ent_yes.mean():.2f}")

print(f"Without Entertainment : ${ent_no.mean():.2f}\n")

print("4. Coach Price: In-flight WiFi (Yes vs No)")

wifi_yes = sample_df[sample_df['inflight_wifi'] == 'Yes']['coach_price']

wifi_no  = sample_df[sample_df['inflight_wifi'] == 'No']['coach_price']

t_stat4, p_value4 = stats.ttest_ind(wifi_yes, wifi_no, equal_var=False)

print(f"T-Statistic    : {t_stat4:.4f}")

print(f"P-Value       : {p_value4:.10f}")

print(f"With WiFi      : ${wifi_yes.mean():.2f}")

print(f"Without WiFi    : ${wifi_no.mean():.2f}")

```

OUTPUT:

=== INDEPENDENT T-TESTS TO COMPARE GROUP MEANS ===

1. Coach Price: Weekend vs Weekday

T-Statistic : 74.2530

P-Value : 0.0000000000

Weekend Mean : \$406.54

Weekday Mean : \$322.32

Price Difference: \$84.22

2. Coach Price: Redeye vs Non-Redeye

T-Statistic : -20.5766

P-Value : 0.0000000000

Redeye Mean : \$321.23

Non-Redeye Mean : \$380.72

3. Coach Price: In-flight Entertainment (Yes vs No)

T-Statistic : 46.1579

P-Value : 0.0000000000

With Entertainment : \$391.52

Without Entertainment : \$320.49

4. Coach Price: In-flight WiFi (Yes vs No)

T-Statistic : 31.8131

P-Value : 0.0000000000

With WiFi : \$384.37

Without WiFi : \$315.14

INTERPRETATION:

All results are highly statistically significant ($p \approx 0$), meaning the differences in coach prices are very strong and reliable.

1. Weekend vs Weekday

- Weekend: \$406.54
- Weekday: \$322.32
- Difference: + \$84.22

Flights are much more expensive on weekends (peak demand pricing).

2. Redeye vs Non-Redeve

- Redeye: \$321.23
- Non-redeye: \$380.72
- Difference: – \$59.49

Redeye flights are cheaper, likely due to lower demand and inconvenience.

3. In-flight Entertainment

- With: \$391.52
- Without: \$320.49
- Difference: + \$71.03

Entertainment is associated with higher-priced (premium) flights.

4. In-flight WiFi

- With: \$384.37
- Without: \$315.14
- Difference: + \$69.23

WiFi is also linked to higher fare classes / premium service.

Z-TEST

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

z_test <- function(x, y) {

  mean1 <- mean(x)

  mean2 <- mean(y)

  sd1 <- sd(x)

  sd2 <- sd(y)

  n1 <- length(x)

  n2 <- length(y)

  z <- (mean1 - mean2) / sqrt((sd1**2/n1) + (sd2**2/n2))

  p_value <- 2 * pnorm(-abs(z))

  cat("Z-Statistic :", round(z, 4), "\n")

  cat("P-Value    :", format.pval(p_value, digits = 10), "\n")

  cat("Redeye Mean :", round(mean1, 2), "\n")

  cat("Non-Redeye Mean :", round(mean2, 2), "\n")

}

z_test(sample_df$coach_price[sample_df$redeye == "Yes"],

        sample_df$coach_price[sample_df$redeye == "No"])
```

OUTPUT:

Z-Statistic : -22.0211

P-Value : < 2.220446e-16

Redeye Mean : 311.27

Non-Redeye Mean : 379.41

INTERPRETATION:

The p-value (< 2.22e-16) is extremely small, showing that the difference in coach prices between redeye and non-redeye flights is statistically significant.

- Redeye Mean: \$311.27
- Non-Redeye Mean: \$379.41

Non-redeye flights are significantly more expensive than redeye flights by about \$68.

The negative Z-statistic (-22.0211) indicates that the mean price of redeye flights is lower than that of non-redeye flights.

PYTHON CODE

```
import pandas as pd

import numpy as np

from scipy import stats

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("==== Z-TEST: REDEYE vs NON-REDEYE FLIGHTS ====\n")

redeye_yes = sample_df[sample_df['redeye'] == 'Yes']['coach_price']

redeye_no = sample_df[sample_df['redeye'] == 'No']['coach_price']
```

```

mean1 = redeye_yes.mean()
mean2 = redeye_no.mean()
std1 = redeye_yes.std()
std2 = redeye_no.std()
n1 = len(redeye_yes)
n2 = len(redeye_no)
z_stat = (mean1 - mean2) / np.sqrt((std1**2 / n1) + (std2**2 / n2))
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
print(f"Z-Statistic    : {z_stat:.4f}")
print(f"P-Value        : {p_value:.10f}")
print(f"Redeye Mean      : ${mean1:.2f}")
print(f"Non-Redeye Mean   : ${mean2:.2f}")
print(f"Sample Size (Redeye)   : {n1}")
print(f"Sample Size (Non-Redeye): {n2}")

```

OUTPUT:

=== Z-TEST: REDEYE vs NON-REDEYE FLIGHTS ===

Z-Statistic : -20.5766

P-Value : 0.0000000000

Redeye Mean : \$321.23

Non-Redeye Mean : \$380.72

Sample Size (Redeye) : 506

Sample Size (Non-Redeye): 9557

INTERPRETATION:

The p-value (0.0000) is far below 0.05, indicating a statistically significant difference in coach prices between redeye and non-redeye flights.

- Redeye Mean Price: \$321.23
- Non-Redeye Mean Price: \$380.72

Non-redeye flights are approximately \$59.49 more expensive than redeye flights.

The negative Z-statistic (-20.5766) shows that redeye flights have a lower average price compared to non-redeye flights.

CHI-SQUARE

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("=== CHI-SQUARE TESTS ===\n\n")

cat("1. In-flight Entertainment vs Weekend\n")

print(chisq.test(table(sample_df$inflight_entertainment, sample_df$weekend)))

cat("\n2. In-flight WiFi vs Weekend\n")

print(chisq.test(table(sample_df$inflight_wifi, sample_df$weekend)))

cat("\n3. Redeye vs Weekend\n")

print(chisq.test(table(sample_df$redeye, sample_df$weekend)))

cat("\n4. In-flight Meal vs Weekend\n")

print(chisq.test(table(sample_df$inflight_meal, sample_df$weekend)))
```

OUTPUT:

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(sample_df$inflight_entertainment, sample_df$weekend)
```

```
X-squared = 0.13575, df = 1, p-value = 0.7125
```

```
data: table(sample_df$inflight_wifi, sample_df$weekend)
```

```
X-squared = 0.91504, df = 1, p-value = 0.3388
```

```
data: table(sample_df$redeye, sample_df$weekend)
```

```
X-squared = 0.00061334, df = 1, p-value = 0.9802
```

```
data: table(sample_df$inflight_meal, sample_df$weekend)
```

```
X-squared = 3.0367, df = 1, p-value = 0.0814
```

INTERPRETATION:

1. In-flight Entertainment vs Weekend

- p-value = 0.7125

No significant relationship exists between in-flight entertainment and whether the flight is on a weekend or weekday.

2. In-flight WiFi vs Weekend

- p-value = 0.3388

No significant association exists between WiFi availability and weekend flights.

3. Redeye vs Weekend

- p-value = 0.9802

Redeye flights are distributed similarly across weekends and weekdays; there is no relationship.

4. In-flight Meal vs Weekend

- p-value = 0.0814

There is no statistically significant association at the 5% level, although the result is slightly closer to significance compared to the other tests.

PYTHON CODE

```
import pandas as pd

from scipy.stats import chi2_contingency

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== CHI-SQUARE TESTS OF INDEPENDENCE ===\n")

print("1. In-flight Entertainment vs Weekend")

cont1 = pd.crosstab(sample_df['inflight_entertainment'], sample_df['weekend'])

chi2, p, dof, expected = chi2_contingency(cont1)

print(f"Chi2 Statistic : {chi2:.4f}")

print(f"P-value      : {p:.10f}")

print(f"Degrees of Freedom: {dof}\n")

print("2. In-flight WiFi vs Weekend")

cont2 = pd.crosstab(sample_df['inflight_wifi'], sample_df['weekend'])

chi2, p, dof, expected = chi2_contingency(cont2)

print(f"Chi2 Statistic : {chi2:.4f}")

print(f"P-value      : {p:.10f}")

print(f"Degrees of Freedom: {dof}\n")

print("3. Redeye vs Weekend")

cont3 = pd.crosstab(sample_df['redeye'], sample_df['weekend'])
```

```
chi2, p, dof, expected = chi2_contingency(cont3)
print(f'Chi2 Statistic : {chi2:.4f}')
print(f'P-value      : {p:.10f}')
print(f'Degrees of Freedom: {dof}\n')
print("4. In-flight Meal vs Weekend")
cont4 = pd.crosstab(sample_df['inflight_meal'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont4)
print(f'Chi2 Statistic : {chi2:.4f}')
print(f'P-value      : {p:.10f}')
```

OUTPUT:

=== CHI-SQUARE TESTS OF INDEPENDENCE ===

1. In-flight Entertainment vs Weekend

Chi2 Statistic : 2.7028

P-value : 0.1001746840

Degrees of Freedom: 1

2. In-flight WiFi vs Weekend

Chi2 Statistic : 0.0461

P-value : 0.8299171805

Degrees of Freedom: 1

3. Redeye vs Weekend

Chi2 Statistic : 2.2572

P-value : 0.1329981133

Degrees of Freedom: 1

4. In-flight Meal vs Weekend

Chi2 Statistic : 0.8396

P-value : 0.3595119067

INTERPRETATION:

1. In-flight Entertainment vs Weekend

- p-value = 0.1002

No significant association exists between in-flight entertainment and weekend/weekday flights.

2. In-flight WiFi vs Weekend

- p-value = 0.8299

WiFi availability is independent of whether the flight is on a weekend or weekday.

3. Redeye vs Weekend

- p-value = 0.1330

No significant relationship exists between redeye flights and weekend status.

4. In-flight Meal vs Weekend

- p-value = 0.3595

Meal availability does not significantly differ between weekend and weekday flights.

PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY
FLIGHTS ===\n\n")

summary_stats <- sample_df %>%

  group_by(weekend) %>%

  summarise(

    Count = n(),

    Mean_Price = mean(coach_price),

    Median_Price = median(coach_price),

    SD = sd(coach_price)

  )

print(summary_stats)

diff <- summary_stats$Mean_Price[summary_stats$weekend == "Yes"] -

  summary_stats$Mean_Price[summary_stats$weekend == "No"]

cat("\nAbsolute Price Difference (Weekend - Weekday) :", round(diff, 2), "\n")

cat("Percentage Increase on Weekend      :", round((diff /

summary_stats$Mean_Price[summary_stats$weekend == "No"] * 100), 2),

"%\n")

t.test(coach_price ~ weekend, data = sample_df, var.equal = FALSE)
```

OUTPUT:

=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===

weekend Count Mean_Price Median_Price SD

<chr> <int> <dbl> <dbl> <dbl>

1 No 3571 322. 327. 54.4

2 Yes 6492 406. 410. 54.7

Absolute Price Difference (Weekend - Weekday) : 83.72

Percentage Increase on Weekend : 25.99 %

Welch Two Sample t-test

data: coach_price by weekend

t = -73.738, df = 7384.7, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-85.94482 -81.49353

sample estimates:

mean in group No mean in group Yes

322.1248 405.8440

INTERPRETATION:

The analysis shows a substantial difference in coach prices between weekend and weekday flights.

- The **average weekday coach price** was **\$322.12**, while the **average weekend coach price** was **\$405.84**.

- This represents an **absolute increase of \$83.72** on weekends.
- Weekend prices were approximately **25.99% higher** than weekday prices.

The Welch two-sample t-test confirmed that this difference is **statistically significant** ($t = -73.738$, $p < 0.001$). The 95% confidence interval indicates that the true mean price difference lies between approximately **\$81.49 and \$85.94**.

Overall, the results suggest that airlines charge significantly higher prices for weekend flights, likely due to increased travel demand during weekends.

PYTHON CODE:

```
import pandas as pd

from scipy import stats

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']

weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']

print("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===\n")

print(f"Number of Weekend Flights : {len(weekend)}")

print(f"Number of Weekday Flights : {len(weekday)}")

print(f"Weekend Mean Price : ${weekend.mean():.2f}")

print(f"Weekday Mean Price : ${weekday.mean():.2f}")

print(f"Absolute Difference : ${weekend.mean() - weekday.mean():.2f}")
```

```
print(f'Percentage Increase      : {((weekend.mean() - weekday.mean()) /  
weekday.mean() * 100):.2f}%")
```

```
t_stat, p_value = stats.ttest_ind(weekend, weekday, equal_var=False)
```

```
print(f'\nt-statistic          : {t_stat:.4f}")
```

```
print(f'p-value              : {p_value:.10f}")
```

OUTPUT:

=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===

Number of Weekend Flights : 6620

Number of Weekday Flights : 3443

Weekend Mean Price : \$406.54

Weekday Mean Price : \$322.32

Absolute Difference : \$84.22

Percentage Increase : 26.13%

t-statistic : 74.2530

p-value : 0.0000000000

INTERPRETATION:

The analysis indicates a clear difference in coach ticket prices between weekend and weekday flights.

- The **average weekend flight price** was **\$406.54**, whereas the **average weekday flight price** was **\$322.32**.
- This shows an **absolute price increase of \$84.22** for weekend flights.
- Weekend fares were approximately **26.13% higher** than weekday fares.

The independent t-test showed that this difference is **statistically significant** ($t = 74.2530$, $p < 0.001$), meaning the variation in prices is unlikely due to random chance.

Overall, the findings suggest that airlines charge significantly higher prices on weekends, likely because of increased passenger demand during peak travel periods.

CORRELATION ANALYSIS AMONG NUMERICAL VARIABLE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
numeric_cols <- c('miles', 'passengers', 'delay', 'hours',
                  'coach_price', 'firstclass_price')
corr_matrix <- cor(sample_df[numeric_cols])
cat("=== CORRELATION MATRIX ===\n")
print(round(corr_matrix, 4))
cat("\n=== Correlation with Coach Price ===\n")
print(round(sort(corr_matrix[, "coach_price"], decreasing = TRUE), 4))
```

OUTPUT:

=== CORRELATION MATRIX ===

	miles	passengers	delay	hours	coach_price	firstclass_price
miles	1.0000	-0.0050	0.0121	0.9850	0.3381	0.2772
passengers	-0.0050	1.0000	-0.0062	-0.0036	0.1720	0.0194
delay	0.0121	-0.0062	1.0000	0.0115	0.0166	0.0054
hours	0.9850	-0.0036	0.0115	1.0000	0.3317	0.2713


```
coach_price    0.3381   0.1720 0.0166 0.3317   1.0000   0.7556
firstclass_price 0.2772   0.0194 0.0054 0.2713   0.7556   1.0000
```

```
> cat("\n=== Correlation with Coach Price ===\n")
```

```
=== Correlation with Coach Price ===
```

```
> print(round(sort(corr_matrix[, "coach_price"], decreasing = TRUE), 4))
```

```
coach_price firstclass_price      miles      hours      passengers
1.0000      0.7556      0.3381      0.3317      0.1720
delay
0.0166
```

INTERPRETATION:

The correlation matrix shows the strength and direction of relationships between numerical variables.

Correlation with Coach Price

- **First-class price ($r = 0.7556$)** has the strongest positive correlation with coach price.

As first-class ticket prices increase, coach prices also tend to increase significantly.

- **Miles ($r = 0.3381$)** and **hours ($r = 0.3317$)** show moderate positive correlations with coach price.

Longer flights and greater travel distances are generally associated with higher coach fares.

- **Passengers ($r = 0.1720$)** has a weak positive relationship with coach price.

Flights with more passengers tend to have slightly higher prices.

- **Delay ($r = 0.0166$)** has almost no correlation with coach price.
Flight delays do not appear to meaningfully affect ticket prices.

Additional Insight

- **Miles and hours** are extremely highly correlated ($r = 0.9850$).
This indicates possible multicollinearity since longer distances naturally require longer flight durations.

PYTHON CODE:

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

numeric_cols = ['miles', 'passengers', 'delay', 'hours',
                'coach_price', 'firstclass_price']

corr_matrix = sample_df[numeric_cols].corr()

print("=== CORRELATION MATRIX ===\n")

print(corr_matrix.round(4))

print("\n=== Correlation with Coach Price (Sorted) ===")

print(corr_matrix['coach_price'].sort_values(ascending=False).round(4))
```

OUTPUT:

=== CORRELATION MATRIX ===

	miles	passengers	delay	hours	coach_price \
miles	1.0000	-0.0059	0.0149	0.9858	0.3408
passengers	-0.0059	1.0000	0.0025	-0.0071	0.1667
delay	0.0149	0.0025	1.0000	0.0192	0.0129
hours	0.9858	-0.0071	0.0192	1.0000	0.3344
coach_price	0.3408	0.1667	0.0129	0.3344	1.0000
firstclass_price	0.2887	0.0106	0.0117	0.2863	0.7547

	firstclass_price
miles	0.2887
passengers	0.0106
delay	0.0117
hours	0.2863
coach_price	0.7547
firstclass_price	1.0000

=== Correlation with Coach Price (Sorted) ===

coach_price	1.0000
firstclass_price	0.7547
miles	0.3408

hours	0.3344
passengers	0.1667
delay	0.0129

INTERPRETATION:

The correlation analysis measures the strength and direction of relationships between variables.

Correlation with Coach Price

- **First-class price** has the strongest positive correlation with coach price ($r = 0.7547$).
Higher first-class fares are strongly associated with higher coach fares.
- **Miles** ($r = 0.3408$) and **hours** ($r = 0.3344$) show moderate positive correlations with coach price.
Longer flights and greater travel distances generally lead to higher coach ticket prices.
- **Passengers** has a weak positive correlation with coach price ($r = 0.1667$).
Flights with more passengers tend to have slightly higher prices.
- **Delay** shows almost no correlation with coach price ($r = 0.0129$).
Flight delays do not significantly influence ticket prices.

Additional Observation

- **Miles and hours** are extremely highly correlated ($r = 0.9858$).
This suggests strong multicollinearity because longer distances naturally require longer flight durations.

LINEAR REGRESSION:

R CODE:

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

model <- lm(coach_price ~ miles + hours + passengers + delay, data =
sample_df)

summary(model)

confint(model)
```

OUTPUT:

```
lm(formula = coach_price ~ miles + hours + passengers + delay,
    data = sample_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-249.361	-42.166	3.179	48.748	157.835

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	134.122770	10.313275	13.005	< 2e-16 ***
miles	0.028505	0.003883	7.342	2.27e-13 ***
hours	-2.153935	2.105736	-1.023	0.306
passengers	0.925243	0.049078	18.852	< 2e-16 ***
delay	0.027095	0.018414	1.471	0.141

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 62.62 on 10058 degrees of freedom

Multiple R-squared: 0.1448, Adjusted R-squared: 0.1444

F-statistic: 425.7 on 4 and 10058 DF, p-value: < 2.2e-16

> confint(model)

2.5 % 97.5 %

(Intercept) 113.906689860 154.33884960

miles 0.020894530 0.03611557

hours -6.281597236 1.97372814

passengers 0.829039697 1.02144670

delay -0.009000196 0.06319060

INTERPRETATION:

he regression model is statistically significant overall ($p < 0.001$), but it explains a modest amount of variation in coach prices ($R^2 = 0.1448$), meaning about **14.5% of price variation** is captured by the model.

Key Findings

- **Miles ($\beta = 0.0285$, $p < 0.001$)**

Significant positive effect: longer flights cost more.

- **Passengers ($\beta = 0.9252$, $p < 0.001$)**

Strong positive effect: more passengers are associated with higher coach prices.

- **Hours ($\beta = -2.15$, $p = 0.306$)**

Not significant after controlling for other variables.

- **Delay ($\beta = 0.0271$, $p = 0.141$)**

Not a significant predictor of price.

PYTHON CODE:

```
import pandas as pd

import statsmodels.api as sm

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

X = sample_df[['miles', 'hours', 'passengers', 'delay']]

X = sm.add_constant(X)    # Add intercept

y = sample_df['coach_price']

model = sm.OLS(y, X).fit()

print(model.summary())
```

OUTPUT:

OLS Regression Results

```
=====
=====
```

Dep. Variable:	coach_price	R-squared:	0.145
Model:	OLS	Adj. R-squared:	0.144
Method:	Least Squares	F-statistic:	425.6
Date:	Sun, 17 May 2026	Prob (F-statistic):	0.00
Time:	23:12:56	Log-Likelihood:	-55838.
No. Observations:	10063	AIC:	1.117e+05
Df Residuals:	10058	BIC:	1.117e+05

Df Model: 4
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
const	145.7542	10.113	14.413	0.000	125.931	165.578
miles	0.0278	0.004	7.108	0.000	0.020	0.035
hours	-1.8957	2.119	-0.895	0.371	-6.049	2.258
passengers	0.8799	0.048	18.289	0.000	0.786	0.974
delay	0.0128	0.015	0.828	0.408	-0.018	0.043
Omnibus:	291.071	Durbin-Watson:		1.994		
Prob(Omnibus):	0.000	Jarque-Bera (JB):		302.594		
Skew:	-0.406	Prob(JB):		1.96e-66		
Kurtosis:	2.747	Cond. No.		3.64e+04		

INTERPRETATION: The regression model is statistically significant overall ($F = 425.6, p < 0.001$), but has modest explanatory power with $R^2 = 0.145$. This means the model explains about 14.5% of variation in coach prices.

Miles

- **Coefficient = 0.0278, $p < 0.001$**

Significant positive relationship: longer distances increase coach prices.

Passengers

- **Coefficient = 0.8799, $p < 0.001$**

Strong and significant effect: more passengers are associated with higher prices.

Hours

- **Coefficient = -1.8957, $p = 0.371$**

Not statistically significant; flight duration does not independently affect price.

Delay

- **Coefficient = 0.0128, $p = 0.408$**

Not significant; delays have no meaningful effect on pricing.

LOGISTIC RREGRESSION

R CODE

```
df <- read.csv("flight_data.csv")  
  
set.seed(10063)  
  
sample_df <- df[sample(nrow(df), 10063), ]  
  
sample_df$redeye_num <- ifelse(sample_df$redeye == "Yes", 1, 0)  
  
sample_df$weekend_num <- ifelse(sample_df$weekend == "Yes", 1, 0)  
  
model <- glm(redeye_num ~ miles + hours + passengers + delay +  
weekend_num,  
             data = sample_df,  
             family = binomial(link = "logit"))  
  
summary(model)
```

```
cat("\n--- CORRECTED ODDS RATIOS & 95% CI (WALD) ---\n")
odds_ratios <- exp(coef(model))
conf_intervals <- exp(confint.default(model))
results_table <- cbind(Odds_Ratio = odds_ratios, conf_intervals)
print(results_table)
```

OUTPUT:

```
summary(model)
```

Call:

```
glm(formula = redeye_num ~ miles + hours + passengers + delay +
     weekend_num, family = binomial(link = "logit"), data = sample_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	169.298103	36.820288	4.598	4.27e-06 ***
miles	-0.004031	0.003532	-1.141	0.2538
hours	2.028547	1.919584	1.057	0.2906
passengers	-0.929865	0.202100	-4.601	4.20e-06 ***
delay	0.001376	0.009503	0.145	0.8849
weekend_num	1.803655	1.084327	1.663	0.0962 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3881.769 on 10062 degrees of freedom

Residual deviance: 36.945 on 10057 degrees of freedom

AIC: 48.945

Number of Fisher Scoring iterations: 14

```
print(results_table)
```

	Odds_Ratio	2.5 %	97.5 %
(Intercept)	3.351442e+73	1.526673e+42	7.357283e+104
miles	9.959775e-01	9.891065e-01	1.002896e+00
hours	7.603032e+00	1.766177e-01	3.272950e+02
passengers	3.946070e-01	2.655442e-01	5.863985e-01
delay	1.001377e+00	9.828993e-01	1.020202e+00
weekend_num	6.071797e+00	7.249968e-01	5.085087e+01

INTERPRETATION:

This logistic regression model predicts the probability that a flight is a **red-eye flight** (`redeye_num = 1`).

The results show that **number of passengers is the only strong and statistically significant predictor**. Its odds ratio ($OR = 0.395$, $p < 0.001$) indicates that as passengers increase, the odds of a flight being a red-eye decrease significantly—by about **60% per unit increase in passengers**.

The variable **weekend_num** suggests that flights on weekends may be more likely to be red-eye flights ($OR = 6.07$), meaning the odds are higher compared

to weekdays. However, this effect is only marginally significant ($p \approx 0.096$) and has a very wide confidence interval, so the result is uncertain.

The variables **miles, hours, and delay** are not statistically significant, indicating they do not have a reliable effect on whether a flight is a red-eye in this dataset.

Overall, the model suggests that **passenger count is the dominant factor**, while weekend timing may have a weak influence, and other predictors do not meaningfully contribute.

PYTHON CODE:

```
import pandas as pd

import numpy as np

import statsmodels.api as sm

import statsmodels.formula.api as smf

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

sample_df['redeye_num'] = sample_df['redeye'].map({'Yes': 1, 'No': 0})

sample_df['weekend_num'] = sample_df['weekend'].map({'Yes': 1, 'No': 0})

formula = 'redeye_num ~ miles + hours + passengers + delay + weekend_num'

model = smf.logit(formula, data=sample_df).fit()

print(model.summary())

print("\n" + "="*20 + " ODDS RATIOS & 95% CI " + "="*20)

params = model.params

conf_int = model.conf_int()
```

```

results_df = pd.concat([params, conf_int], axis=1)
results_df.columns = ['Odds Ratio', '2.5%', '97.5%']
results_df = np.exp(results_df)
print(results_df)

```

OUTPUT:

Optimization terminated successfully.

Current function value: 0.000753

Iterations 19

Logit Regression Results

```

=====
=====

```

```

Dep. Variable:      redeye_num  No. Observations:      10063
Model:              Logit  Df Residuals:      10057
Method:             MLE  Df Model:      5
Date:              Mon, 18 May 2026  Pseudo R-squ.:      0.9962
Time:              02:07:53  Log-Likelihood:      -7.5730
converged:          True  LL-Null:      -2006.0
Covariance Type:    nonrobust  LLR p-value:      0.000

```

```

=====
=====

```

```

      coef  std err      z  P>|z|  [0.025  0.975]

```

```

-----

```

```

Intercept  244.1917  84.365   2.894   0.004   78.838  409.545
miles      -0.0010   0.007  -0.140   0.889  -0.015   0.013

```

hours	0.5711	3.474	0.164	0.869	-6.238	7.380
passengers	-1.3396	0.460	-2.912	0.004	-2.241	-0.438
delay	-0.0526	0.113	-0.466	0.641	-0.274	0.169
weekend_num	0.1644	1.474	0.112	0.911	-2.725	3.054

=====
 =====

Possibly complete quasi-separation: A fraction 0.99 of observations can be perfectly predicted. This might indicate that there is complete quasi-separation. In this case some parameters will not be identified.

===== ODDS RATIOS & 95% CI
 =====

	Odds Ratio	2.5%	97.5%
Intercept	1.124914e+106	1.734145e+34	7.297151e+177
miles	9.990164e-01	9.852998e-01	1.012924e+00
hours	1.770280e+00	1.954395e-03	1.603511e+03
passengers	2.619454e-01	1.063216e-01	6.453569e-01
delay	9.488051e-01	7.606280e-01	1.183536e+00
weekend_num	1.178727e+00	6.555550e-02	2.119420e+01

INTERPRETATION:

The model indicates that **passengers is the only statistically significant predictor** of whether a flight is a red-eye.

- **Passengers (coef = -1.34, OR $\approx$ 0.26, p = 0.004):**

A higher number of passengers significantly **reduces the odds of a red-eye flight**. Specifically, each unit increase in passengers is associated with about a **74% decrease in odds** (since $1 - 0.26 \approx 0.74$). This is a strong and reliable effect.

Miles, hours, delay, weekend_num:

All have **p-values > 0.05**, meaning there is no statistically reliable evidence that they affect red-eye probability in this model. Their confidence intervals are also wide, showing high uncertainty.